NEW ZEALAND SPORTS ECOSYSTEM

AI Landscape, Stakeholder Intelligence & Innovation Roadmap

A Consolidated Research Report for Rapid Build Immersion

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Executive Summary

The New Zealand sports ecosystem operates within a highly distinct matrix characterised by geographical isolation, a reliance on grassroots volunteerism, strict bicultural obligations under Te Tiriti o Waitangi, and a structural dependence on fluctuating community funding models. This consolidated report evaluates the strategic priorities of a 16-member delegation representing the apex of New Zealand's sports governance, high-performance infrastructure, broadcasting technology, and diplomatic sectors.

The ecosystem is under immense pressure. Community sports face critical funding shortfalls driven by declining Class 4 gaming revenues, while high-performance entities struggle to maintain global competitiveness against better-resourced nations amid escalating travel and logistical costs. Participation among rangatahi (ages 12-17) has declined since 2021, even as newer sports like basketball experience explosive growth. The sector generates an estimated NZ$20.8 billion in total socio-economic value — yet digital maturity remains highly fragmented across codes.

The global AI in sport market reached USD$7.63 billion in 2025 and is projected to reach USD$27 billion by 2030. Yet NZ's adoption is uneven: world-leading in niches like America's Cup design (ETNZ's RL-powered digital twin), broadcast graphics (Virtual Eye/ARL), and prosthetics development (Zenith Tecnica/HPSNZ), but nascent across community sport, governance, and commercial operations.

This report delivers four integrated outputs:

- Deep profiles of all 16 delegation members, with strategic priorities, pain points, and technology interests
- A ten-dimension analysis of NZ's sports ecosystem, mapping structural strengths and systemic gaps
- A comprehensive audit of 60+ existing AI tools across 14 categories globally, with NZ adoption realism ratings
- Sixteen novel, unprecedented AI application proposals designed specifically for NZ's resource-constrained, multi-code, bicultural context — none replicable by importing off-the-shelf global products

The report's overarching finding: NZ's small scale, which is typically a constraint, becomes a decisive advantage when it enables whole-of-system AI coordination that larger, more fragmented nations cannot achieve. The strategic imperative is not whether NZ sport adopts AI — it is whether NZ moves fast enough to leverage its centralisation advantage before better-resourced competitors widen the gap.

Section 1 — Delegate Profiles

1.1 Government & Policy

Hon Chris Bishop — Associate Minister for Sport and Recreation

Chris Bishop is a National Party MP responsible for Vote Sport and Recreation, managing an appropriation of approximately NZ$122 million for the 2024/25 fiscal year. The portfolio encompasses Sport NZ (community sport leadership), High Performance Sport NZ, grants to NSOs, Active Schools programmes, and the newly established Sport Integrity Commission Te Kahu Raunui. Bishop is a lawyer by training and career politician, operating under a strict fiscal discipline mandate.

Budget Oversight: Crown allocation of ~NZ$32.1m to Sport NZ plus ~NZ$74.2m to HPSNZ, with ~NZ$82m/year from lottery grants. He commissioned an independent rapid review of Sport NZ in late 2024, stressing accountability, efficiency, and value-for-money.

Sport Diplomacy: Launched NZ's first Sport Diplomacy Strategy (2025-2030), coordinating nine government agencies. Joined the PM's trade mission to India, signing an MOU between Sport NZ and India's Youth & Sports Ministry, framing sport as a 'universal language' that builds people-to-people links supporting trade.

Socio-Economic Framing: Physical activity generates an estimated NZ$20.8 billion in total value — NZ$9.02 billion in social value, NZ$3.09 billion in physical health benefits, and NZ$3.96 billion in economic value. Bishop requires empirical evidence that technology investments yield quantifiable returns against this benchmark.

Technology Interest: Data analytics demonstrating value-for-money; digital tools for governance and compliance; outcome-based ROI modelling for sport investment.

Key Tension: Balancing elite high-performance Olympic/Paralympic campaigns against the critical need to sustain community sport infrastructure under fiscal constraint.

Peter Miskimmin QSO — Sports Diplomacy Manager, NZ Government

Peter Miskimmin's career is foundational to the modern NZ sports system. A former Black Sticks field hockey captain who represented NZ at the 1984 Olympics, he served as Chief Executive of Sport NZ from 2008 to 2020 — the longest-serving CEO in the organisation's history, overseeing record-breaking Olympic eras. Awarded the Companion of the Queen's Service Order (QSO), he left a legacy of systemic reforms addressing declining youth participation and equity for marginalised demographics.

Current Role: Cross-government Sports Diplomacy Manager reporting through MFAT and NZ Trade & Enterprise. Tasked with using sport to achieve government objectives across political, economic, and social sectors — coordinating international sporting engagements as soft-power assets.

Strategic Priorities: Quantifying the geopolitical and economic return on international sporting engagements; leveraging NZ's sporting brand for cultural diplomacy in Asia-Pacific; connecting sport performance to trade and bilateral relationship outcomes.

Technology Interest: Platforms connecting sport and trade data; AI tools that can empirically measure the diplomatic ROI of sports tours and exchanges.

1.2 National Sport System Leadership

Raelene Castle ONZM — Group Chief Executive, Sport NZ & HPSNZ

Raelene Castle brings formidable commercial and governance experience from tenures as CEO of Rugby Australia (2017-2020), CEO of Canterbury-Bankstown Bulldogs NRL (2013-2017), and CEO of Netball NZ (2007-2013) — making her the first female CEO of an Australian professional sports club. Appointed Group CE of Sport NZ and HPSNZ in December 2020, she oversees the full national sports leadership body including ~NZ$114m/year in Crown and lottery funding across 44 sports.

Every Body Active Strategy: Castle's flagship 2024-28 strategy targets five outcomes: (1) increased weekly activity for tamariki (children 5-11), (2) increased weekly activity for rangatahi (teens 12-18), (3) better quality sport experiences, (4) more equitable access for women, Māori, Pacific peoples, and disabled communities, and (5) strong system-wide leadership.

Bicultural Integration: Castle has championed unprecedented integration of Te Tiriti o Waitangi frameworks into both high-performance and community systems, embedding the concepts of Ihi (personal energy), Wehi (emotional response), and Wana (collective energy) as strategic pillars.

Key Challenges: Reversing rangatahi dropout rates; ensuring both elite medal generation and widespread wellbeing are measurable; managing a 'whole-of-sport' system where community and elite outcomes reinforce each other.

Technology Interest: Data analytics for participation monitoring (ActiveNZ); digital engagement campaigns; evidence-based decisions on funding allocation; AI tools that bridge grassroots wellbeing with elite performance data.

Belinda Edwards — Chief Executive, Basketball New Zealand

The first female CEO of Basketball New Zealand (appointed permanently in October 2025), Edwards leads a sport experiencing explosive demographic growth — basketball is now the second most popular secondary school sport in NZ, with 107,444 registered participants across 36 regional associations and a 93.5% growth in female participation. Despite this surge, BBNZ faces severe funding constraints: Junior Tall Blacks players were recently forced to self-fund NZ$5,000 each to attend the U19 World Cup.

Strategic Pillars: Participation Impact, International Performance, Fan Experience, and Diversity. The 'KiwiHoops' grassroots programme and Tall Ferns/Tall Blacks pathways are core delivery vehicles.

Technology Priority: Cohesive community sport participation platforms to reduce administrative burden on volunteer-run regional associations; social media and digital fan engagement to build commercial base; analytics for talent pipeline and scouting.

Key Pain Point: Funding constraints that force athlete self-funding; lack of coordination tools for 36 disparate regional associations; limited broadcast economics for tier-two sports.

Jeff Latch — Chief Executive, Golf New Zealand

Jeff Latch joined Golf NZ as CEO in March 2024, having previously served as Spark Sport's Chief Business Officer and a television sports producer. He oversees a sport managing a post-COVID participation boom — adult club memberships rose ~4.6% in 2023 with over 200,000 casual players (35% of rounds now played by women) — while simultaneously navigating existential threats to rural clubs from aging demographics and high operational costs.

Strategic Plan: 'Enriching Lives Through Golf' (2020-2025): growing the game's profile, increasing participation especially among women and juniors, and sustaining financially fragile rural clubs through the 'OnCourse New Zealand' digital sustainability programme with the GEO Foundation.

Notable Asset: DotGolf, founded in Auckland in 1998, created the world's first national centralised handicapping system and is jointly owned 50/50 by The R&A and Golf NZ — serving as official technology provider for England Golf, Wales Golf, Golf Ireland, Scottish Golf, and Golf Australia. This is an underutilised AI platform opportunity.

Technology Interest: Handicap and booking system improvements; digital coaching aids (swing analysis apps like TrackMan, Arccos); environmental optimisation technology for water and energy use at rural facilities.

Cam Mitchell — Chief Executive, Athletics New Zealand

Appointed in August 2023 after successful stints as CEO of Cricket Wellington and regional football bodies, Mitchell brings a commercial and facility-management pedigree to Athletics NZ. His mandate bridges the high-performance pipeline for Olympic and Paralympic campaigns with foundational grassroots needs. Athletics NZ supports 80+ clubs nationwide and produced Olympic medallists including Valerie Adams and Zane Robertson.

Strategic Focus: Increasing youth (12-18) retention; introducing flexible membership; improving events quality; ensuring high-performance funding continuity through Paris 2024 cycle and beyond to LA 2028.

Infrastructure Challenge: Many NZ tracks are aging with no indoor facilities; funding and access across varying regional jurisdictions is inconsistent; competition from other sports for talent.

Technology Interest: Performance tracking systems; open athletic databases for coaches; facility planning tools; AI-powered athlete identification from school physical education data.

Michelle Hollands — Chief Executive, Hockey New Zealand

Appointed in late 2024, Hollands is a former Black Stick with 99 international caps and an Olympian (1992). Her corporate consulting firm Freshminds gave her deep operational acumen that she brought to previous work designing Hockey NZ's pathways project (Hiki Tu) and the 2021 high-performance review (Project Kahurangi). She leads an organisation that co-hosted the 2022 Women's Hockey World Cup and whose Black Sticks teams remain globally competitive.

Strategic Priorities: Aligning community and elite tiers; diversifying revenue beyond traditional grants; expanding turf access including planning a new facility at Ōtākaro/Olympic Park in Christchurch; managing complex international event logistics.

Technology Interest: Performance analytics in hockey (tracking passes, positions, tactical patterns); HockeyTV streaming upgrades; Catapult wearables; Pixellot AI camera deployment already underway across NZ domestic hockey via CelloSport partnership.

Graham Parks — Interim Chief Executive, NZ Cricket

Serving as Interim CE following the abrupt departure of Scott Weenink in late 2025, Parks was previously NZC's Chief Venues and Events Officer. He oversees an organisation reporting a NZ$2.212 million financial surplus and historic on-field success — the White Ferns' T20 World Cup victory and the Black Caps' unprecedented test sweep in India. However, Parks inherits significant governance tensions around the potential privatisation of T20 franchise leagues ('Project Underground') and a looming funding shortfall from declining Class 4 gambling revenues.

Commercial Landscape: NZC secured a new six-year broadcast deal with Sky from 2026. Partnerships include Virtual Eye (Innovation Partner), CricViz (seven-year Performance Analysis and Scoring deal), and NZ Cricket was part of PM Luxon's March 2025 India trade delegation. The ICC-Google partnership for T20 World Cup 2026 uses Gemini AI for fan engagement.

Key Challenges: Balancing domestic major associations with commercial expansion; maintaining broadcast relationships; protecting grassroots funding as gambling revenues decline.

1.3 International & Foundation

Suzannah Jessep — Chief Executive, Asia New Zealand Foundation

A career diplomat with postings including NZ High Commissioner to India (2006-2009) and Deputy High Commissioner to Malaysia, Jessep became CEO of the Asia NZ Foundation in November 2023. The Foundation is government-funded to strengthen NZ's ties with Asia through people-to-people exchanges across business, arts, education, and sport. Under Jessep, NZ launched its first Sport Diplomacy Strategy 2025-2030, coordinating nine government agencies with an initial focus on India, Pacific nations, and the United States.

Sport Diplomacy Programmes: Sports Leaders Fund (scholarships for NZ sports professionals in Asia); Sports Connector Programme (bringing Asian sports leaders to NZ); coaching exchanges between NZ and Asian national bodies; sport-for-development projects in the Pacific.

2026 Priority: The India-NZ 'Sporting Unity' centenary year, marking 100 years since the Indian hockey team's 1926 NZ tour, is the flagship diplomatic sports event. India's cricket-tech ecosystem (1,800+ startups, USD$5.7 billion projected market by FY29) offers partnership opportunities in cricket, hockey, and sports science.

Technology Interest: AI tools to measure the diplomatic and economic ROI of sports exchanges; digital platforms connecting NZ-Asia sport communities; real-time translation tools for Hindi, Tamil, Marathi, and Pacific Island languages.

Kirsty Sharp — Manager, Sport Programme, Asia New Zealand Foundation

With 20+ years in NZ sport, Sharp executes the Foundation's sports strategy on the ground. Her roles have spanned leading the 'Rising Legends' initiative in Gisborne and working on the 2017 Rugby League World Cup legacy in NZ. She manages day-to-day administration of exchange programmes, outcomes measurement, and coordination with Peter Miskimmin's office to align government sport diplomacy with AsiaNZ activities.

Focus: Executing sports exchange programmes; outcomes measurement; matching NZ expertise with Asian capacity needs; co-ordinating with NSOs to source participants for Asia tours.

1.4 Paralympic & Inclusive Sport

Maria Clarke — Deputy Chair, Paralympics New Zealand

A leading international sports lawyer and governance expert, Clarke established NZ's first sports law practice and holds board roles across World Sailing, World Athletics, WADA, the IPC, and UCI. Appointed to PNZ's board in 2023 and serving as Deputy Chair, she is also Vice-Chairperson of the IPC's Governance Review Working Group — architecting structural reforms to strengthen the global paralympic movement ahead of LA 2028 and Brisbane 2032.

Strategic Priorities: Securing diverse and sustainable revenue streams for para-sport; amplifying the athlete voice at board level; ensuring objective and equitable classification pathways; expanding accessible grassroots participation through PNZ's five regional hubs.

Key Pain Points: Classification requires expensive international travel for human expert assessment — costly and inequitable for a geographically isolated nation. Accessible facilities and transport remain gaps. Para-sport funding competes with mainstream sports budgets.

Technology Interest: AI tools for remote classification via video analysis; computer vision for accessibility auditing of venues; AI monitoring for athlete welfare and governance consistency.

1.5 Sport Technology & Broadcasting

Ben Taylor — Executive VP, Production & Commercial, Virtual Eye

Ben Taylor oversees Virtual Eye (the sports division of Animation Research Ltd, based in Dunedin), an Emmy Award-winning company that has won four Sports Emmys. Virtual Eye provides real-time 3D graphics, drone integration, predictive ball-tracking, and advanced spatial telemetry for the PGA Tour, MLB, America's Cup, NZ Cricket, Formula 1, and 90+ golf tournaments annually. Their COVID-era innovation — full remote production from Dunedin via AWS cloud — reduced crew deployment by over 80% while maintaining broadcast quality.

2025 Partnerships: Named NZ Cricket's official Innovation Partner (2025) for high-performance analytics and broadcast innovation. The 37th America's Cup in Barcelona (2024) showcased Virtual Eye's WindSight IQ technology using LiDAR sensors for sub-0.1 m/s wind accuracy, creating AR visualisations that let viewers 'see the wind'.

Technology Focus: Predictive ball-tracking; 3D broadcast storytelling; low-latency rendering; biomechanical analysis tools for high-performance programmes; expanding into cloud-based remote production and emerging markets.

Innovation Gap: All current 3D telemetry is purely visual — entirely inaccessible to the visually impaired fan community. This represents a significant accessibility and market opportunity.

1.6 Sport Quality & Integrity

Geoff Allott — Executive Director, Quality New Zealand

A former professional cricketer, Allott champions the implementation of rigorous quality assurance, governance, and compliance standards across the sports ecosystem. His professional focus aligns with the objectives of the Sport Integrity Commission Te Kahu Raunui (launched 1 July 2024), which combined Drug Free Sport NZ with broader integrity functions under the Integrity Sport and Recreation Act 2023 — managing anti-doping, safeguarding, match-fixing, complaints, and dispute resolution on a budget of approximately NZ$10.6 million.

Strategic Focus: Embedding anti-doping, safeguarding, competition manipulation prevention, and transparent organisational protocols efficiently across all tiers of sport. Advocating for the Sport Integrity Code's six minimum standards (including child safeguarding and abuse prevention) as mandatory frameworks for all Crown-funded bodies.

Unique NZ Context: The integrity framework incorporates indigenous values — Whanaungatanga (relationships), Haumarutanga (safety), and Tikanga Māori principles — as core welfare protocols, going beyond standard international models.

1.7 Professional Sport & Athlete Perspective

Ross Taylor — Former NZ Cricketer & Sports Ambassador

One of NZ's most celebrated cricketers, Taylor played for the Black Caps from 2006 to 2022, becoming NZ's all-time leading Test run-scorer with over 7,500 runs across 200+ internationals. Awarded a CNZM in 2022, his autobiography Black and White brought critical, long-overdue attention to systemic racism and cultural diversity issues in elite NZ cricket. He now serves as a Sport NZ Sports Ambassador (2023) and is associated with the Seattle Orcas in Major League Cricket USA.

Advocacy Priorities: Enhanced athlete mental health support; equitable pathways for Māori and Pasifika athletes; modernisation of player welfare frameworks in an increasingly franchised, high-pressure global sports economy. He was part of PM Luxon's March 2025 India delegation in a Sports Ambassador capacity.

Technology Interest: Platforms connecting fans and players; AI tools that monitor athlete wellbeing or offer counselling; digital fan engagement (1M+ social media followers); innovations that protect player privacy while improving sport access and fairness.

David Dome — General Manager, Wellington Phoenix FC

General Manager of Wellington Phoenix since 2022, Dome navigates the unique logistical and commercial challenges of running NZ's only professional football club — operating within Australia's A-League while building a domestic fan base of 10,000+ members. His operational resilience was forged during the pandemic when the Phoenix relocated temporarily to Australia, highlighting the critical importance of digital revenue streams independent of physical attendance.

Strategic Focus: Digital fan engagement across TikTok, YouTube, and bespoke apps (700k+ social media followers); localized marketing; sustaining Sky Sport viewership metrics; commercialising the new NZCIS High Performance Centre as both a training and marketing asset.

Key Challenges: High cross-Tasman travel costs; small NZ market relative to Australian A-League clubs; competition from rugby for talent and broadcast share; exclusion from the Australia Cup creating competitive schedule gaps.

Technology Interest: Digital ticketing and fan personalisation; AI-driven fan apps; GPS player monitoring; analytics from match data (StatsBomb/Opta access via A-League).

Brett O'Riley — Chief Executive, World Bowls

An experienced corporate leader executing an aggressive global strategy to modernise a traditional sport, O'Riley's priorities include positioning bowls as a leading driver for positive active aging (NZ has ~35,000 registered players), establishing a unified World Bowls Series integrating all codes, expanding global digital governance structures, and strategically pursuing inclusion in the 2032 Olympic and Paralympic Games in Brisbane. Global Lawn Bowls announced a partnership with STWS for digital scoring and tournament management in November 2025 — but virtually zero AI applications exist for the sport globally, representing a genuine greenfield opportunity.

Section 2 — New Zealand Sports Ecosystem Analysis

2.1 System Architecture

NZ's sport funding flows hierarchically: government allocates via Vote Sport and Recreation (NZ$122m annually) through Sport NZ and HPSNZ to NSOs, Regional Sports Trusts (RSTs), and local clubs. The high-performance investment framework (Tū te Ihi/Performance Pathways) is targeted toward sports with consistent medal potential, using mechanisms including Prime Minister's Scholarships and tailored campaign investments. NZ's centralised coordination through HPSNZ across 36 NSOs enables unified data strategies and cross-sport technology transfer that would be impossible in fragmented systems — but this advantage is currently underutilised.

NZ's first national AI strategy was released in July 2025 (the last OECD country to do so), explicitly positioning the country as an 'adopter nation' focused on implementing proven solutions. This pragmatic approach aligns well with sport: NZ should adopt cloud-based AI platforms rather than building bespoke systems, except in areas of existing excellence (broadcast graphics, sailing design, prosthetics).

2.2 Participation Trends

NZ faces a stark divergence in participation trends. Sport NZ's Active NZ surveys (2017-2023, 159,000+ respondents) reveal tamariki (5-11) participation rebounding to ~95% active weekly post-COVID, but rangatahi (12-17) activity declining since 2021 — especially severe for ages 13-15. Club membership is around 25% of adults. Emerging sports like basketball (107,000+ registered, 93.5% female participation growth) are surging, while traditional club-based sports like golf and rugby manage aging demographics. Digital disruptors (screen time) drive youth dropout, pushing Sport NZ toward promoting unstructured 'active recreation' alongside formal competition.

2.3 High-Performance Pipeline

HPSNZ's strategy relies on centralised daily training environments and specialist practitioner support. The Goldmine innovation team directly supported 16 of 20 Paris 2024 Olympic medallists. Key partnerships: HPSNZ/Cub Digital for AI-powered video analysis in swimming and athletics; CrewLAB/Rowing NZ for AI-driven talent pathway automation (signed September 2024); Zenith Tecnica (Auckland) for 3D-printed custom titanium prosthetics for Paralympians; AWS NZ for cloud-based performance analytics for NZ Rugby. The pipeline challenge is geographic dispersion of talent across a small population and exorbitant international competition costs — forcing tier-two sport athletes to self-fund development.

2.4 Sport Integrity & Wellbeing

The ecosystem underwent a massive regulatory transformation with the establishment of the Sport Integrity Commission Te Kahu Raunui on 1 July 2024. The Commission operates on ~NZ$10.6 million covering anti-doping, safeguarding, match-fixing, complaints, and dispute resolution. An Integrity Code with six minimum standards is being implemented across all Crown-funded bodies. Uniquely, NZ's framework incorporates Tikanga Māori principles (Whanaungatanga, Haumarutanga) into core welfare protocols. Globally, Cricket saw suspicious matches triple in 2025 — directly relevant for NZC's international profile.

2.5 Digital Maturity

Digital maturity is highly fragmented. Elite broadcasters and high-performance units utilise world-leading technology (Virtual Eye's spatial telemetry, Hawk-Eye ball tracking, Catapult wearables across

NZ Rugby and Black Caps). However, grassroots NSOs rely on disparate entry-level CRMs or manual processes. Enterprise IT providers like Datacom and Spark dominate generic architecture, but sports-specific, cross-code data interoperability is virtually non-existent. NZ's 14 Regional Sports Trusts already use Sport NZ's Power BI-based Insights Tool — a natural deployment point for AI upgrades. Stats Perform/OptaAI Studio provides AI-driven broadcast insights for Sky NZ's rugby broadcasts (real-time match momentum, kick predictions, advanced player statistics).

2.6 Fan Engagement

Fan engagement is shifting from physical attendance to continuous digital touchpoints. Professional franchises like the Wellington Phoenix use dedicated digital teams across TikTok, YouTube, and apps. NZ Rugby launched NZR+ as a digital fan platform. NZ Cricket partnered with Qrious (Spark subsidiary) for data-led fan engagement, overlaying participant data with Census data to identify under-representation. NZ Cricket's CricViz deal provides fan-facing analytics. However, personalisation at scale remains a gap for smaller NSOs. IBM's 2025 global study found 85% of fans see value in AI integration and 82% use mobile apps during events.

2.7 Sport Diplomacy

Geographic isolation makes international competition expensive, but NZ leverages sport deeply as soft power. The Asia NZ Foundation's Sports Connector Programme facilitates cross-border athlete, coach, and executive exchanges to build bilateral trade and cultural relationships. The 2026 India-NZ 'Sporting Unity' centenary, combined with India's booming sports-tech ecosystem and the newly signed Sport MOU, creates an unprecedented corridor for bilateral technology collaboration. NZ's total international sport diplomacy budget remains constrained, with no AI tool currently available to quantify its ROI or optimise deployment.

2.8 Paralympic & Inclusive Sport

Para-sport faces immense equity and logistical hurdles. Classification currently requires athletes to travel internationally to be assessed by human experts — a massive cost burden for a geographically isolated nation. HPSNZ's Goldmine team and Zenith Tecnica's bespoke prosthetics (e.g. for Paralympians Holly Robinson and Anna Grimaldi) are world-leading examples of NZ para-sport innovation. PNZ's 2021-2032 strategy targets both Paris 2024 performance and Brisbane 2032 grassroots expansion. AI pose estimation tools perform poorly on wheelchair users — the landmark WheelPose study (CHI 2024) found no existing AI training datasets including people with disabilities performing sports.

2.9 Commercial Landscape

NZ's commercial model is perilously concentrated on two pillars: broadcast rights and community grants. Sky Sport retains dominance, recently raising subscription prices 10% to cover premium rights costs. NZ Rugby secured a 2026-2030 deal with Sky and TVNZ; NZ Cricket finalised a new six-year Sky deal from 2026; Netball NZ paid ~NZ$600-700k to TVNZ to air games in 2026. A severe threat looms over grassroots from the decline of Class 4 gambling (pokie machine) revenues, driven by regulatory changes and the shift to online wagering. NZ's total sports sponsorship market is approximately NZ$200m/year — requiring NSOs to fiercely compete for limited corporate support.

2.10 Finalised Challenge Framework

The following ten challenges represent the definitive, prioritised set of systemic problems that AI interventions must address. These are the base context for all rapid build sessions and novel application development.

1	Fragmented data limits whole-of-system decisions
Body	New Zealand sport is coordinated at the national level, but operational data remains fragmented across agencies, sports, clubs, and programs. This makes it difficult to make confident, evidence-based decisions on participation, funding, facilities, pathways, integrity, and long-term outcomes across the system.
Proof	• Data sits across multiple organisations, platforms, and reporting structures. • Cross-code interoperability is weak, limiting joined-up planning. • Leaders lack a single, trusted view of system performance.

2	Youth participation drops sharply during the teenage years
Body	Children's participation has recovered more strongly than teenage participation, creating a serious retention challenge at a critical life stage. The issue is no longer only about access to sport, but about keeping young people engaged through adolescence in relevant, flexible, and inclusive ways.
Proof	• Teenage participation remains below stronger child recovery trends. • The decline is especially important for long-term sporting pathways. • Youth re-engagement is central to the national participation agenda.

3	Volunteer-run clubs are stretched by invisible operational work
Body	Community sport depends heavily on volunteers, but too much time is still spent on admin, coordination, scheduling, communications, and compliance. This hidden workload reduces volunteer sustainability, affects club quality, and creates friction at the grassroots level where participation experiences are shaped.
Proof	• Thousands of clubs rely on volunteer-led operating models. • Admin burden directly affects retention of volunteers and organisers. • Grassroots delivery remains less digitised than elite sport environments.

4	Quality coaching is uneven outside major centres
Body	Access to coaching, specialist feedback, and structured development support is not evenly distributed across New Zealand. Rural athletes, smaller sports, and community-level participants often receive less guidance than those connected to major centres or high-performance environments, widening the gap in development quality.
Proof	• Geography and dispersion create access challenges across the country. • Elite support systems are stronger than grassroots coaching access. • Regional and rural communities face weaker development infrastructure.

5	Integrity and safeguarding are still too reactive
Body	New Zealand sport is strengthening its integrity framework, but many risks are still identified after harm has already occurred. The challenge is to move from reactive case management toward earlier visibility across safeguarding, athlete welfare, abuse prevention, and competition-related integrity concerns.
Proof	• Integrity reform is already underway at the national level. • Safeguarding and wellbeing concerns remain live across the ecosystem. • Monitoring and early-warning capability is still uneven across sports.

6	Para-sport pathways still face access and fairness barriers
Body	Para-sport is strategically important, yet athletes still face uneven access to classification, facilities, coaching, and progression opportunities. Classification remains expert-driven and resource-intensive, making fairness, timeliness, and consistency harder to achieve for a geographically dispersed athlete base.
Proof	• Classification processes are still costly and specialist-dependent. • Access varies significantly by location and support availability. • Equity in para pathways is a growing priority toward future Games cycles.

7	Sport diplomacy creates strategic value, but the impact is hard to prove
Body	New Zealand increasingly uses sport as a tool for diplomacy, visibility, and international partnership-building. However, these outcomes are hard to measure in a structured way, making it difficult to compare initiatives, justify investments, and link sporting engagements to trade, tourism, education, or bilateral value.
Proof	• Sport diplomacy is becoming a recognised strategic agenda. • Outcomes span soft power, trade, tourism, and relationships. • Measurement remains fragmented and often anecdotal.

8	Lower-tier sport struggles to convert visibility into sustainable economics
Body	The commercial sports model is uneven. Premium sports attract audiences and rights value, but many smaller, women's, and second-tier competitions struggle to fund coverage, grow audiences, and build sustainable revenue. Visibility exists, but monetisation remains difficult and inconsistent.
Proof	• Commercial strength is concentrated in a few top-tier sports. • Smaller competitions face weaker broadcast and sponsorship economics. • Traditional funding sources are under growing pressure.

9	Funding pressure is forcing sharper trade-offs across the system

Body	The sports ecosystem is being asked to deliver community participation, elite success, inclusion, integrity, and innovation at the same time, but within tight budgets. This creates tension around where funding should go, what outcomes matter most, and how value is demonstrated convincingly.
Proof	• Public funding decisions are under stronger value-for-money scrutiny. • Community and elite priorities compete for finite resources. • Rising operating costs increase pressure on allocation choices.

10	**Digital maturity is highly uneven across the ecosystem**
Body	Some parts of New Zealand sport are globally advanced, particularly in high performance and sports broadcast innovation, while others still operate with basic digital tools and manual workflows. This uneven maturity makes it difficult to scale innovation consistently across the broader sporting system.
Proof	• High-performance and broadcast environments show strong innovation maturity. • Community and mid-tier organisations remain less digitally enabled. • Uneven foundations slow wider adoption of new capabilities.

Section 3 — AI in Sports: Global Landscape Audit

3A — What Already Exists: Comprehensive Audit

The following audit maps known AI tools worldwide across 14 categories. Each entry includes example products, core functionality, and critical limitations relevant to NZ's context. The global AI in sport market is USD$7.63 billion (2025), projected to reach USD$27 billion by 2030. Understanding what exists is essential to ensure subsequent proposals are genuinely novel.

Category	Key Tools & Capabilities	Critical Limitations / Gaps
Athlete Performance & Training	Catapult Sports (Australia, FIFA-approved, 1,200+ football teams globally); STATSports (one European client cut soft-tissue injuries from 44 to 22 in Year 1); Zone7 (50+ clubs globally, 72.4% accuracy, Getafe 40% injury reduction Year 1, 66% Year 2); Kitman Labs (Chelsea, Manchester United — 30% fewer injuries, 10% higher availability)	Elite-only; physiological baselines Euro/US-centric; cost-prohibitive for amateur tiers; siloed per sport. No algorithmic recalibration for indigenous physiologies (Māori/Pasifika).
Talent ID & Scouting	aiScout; Comparisonator; CrewLAB (partnership with Rowing NZ, September 2024); Stats Perform Scout; Ajax AI video (70% scouting time reduction, 45% accuracy increase); Intel/IOC pilot (Paris 2024 — 1,000+ youth screened, 48 identified in Senegal)	Fiercely sport-specific; fail to detect transferable biometrics across codes; no cross-code talent fluidity for small nations.
Fan Engagement	WSC Sports (530+ clients, 10.2M highlights in 2024 — 1 every 3.1 seconds; NBA app engagement tripled, video views up 700%); IBM Watson; GameDay (NZ/AU, launched AI chatbot 2024); Waresport AI Copilot; Golden State Warriors' AI demand forecasting (27% revenue increase, 92% accuracy, 50+ variables)	Generic text-based chatbots; no cross-sport fan personalisation; no accessibility for disabled fans; small NZ market limits dynamic pricing applicability.
Broadcasting & Media	Virtual Eye/ARL (Dunedin — 4 Sports Emmys, PGA Tour, MLB, America's Cup, NZ Cricket); Hawk-Eye (Sony, 550+ staff, 23 of top 25 sports leagues, SkeleTRACK 29 skeletal points); Pixellot (30,000+ systems, 5M+ games; deployed in NZ for hockey via CelloSport across North Harbour, Wellington, Manawatu, Waikato; Netball NZ confirmed early adopter 2026); WSC Sports; Stats Perform/Opta (Sky NZ rugby broadcasts); TrackMan AI Motion Analysis V2 (2025)	World-leading visual rendering; entirely lack non-visual sensory translation (haptics, spatial audio for visually impaired fans). Community sport largely lacks any broadcast coverage.
Governance & Integrity	Sportradar (monitors 1M+ events across 70 sports; AI-flagged suspicious matches up 56% YoY to 682 in 2025); WADA ADAMS analytics/ABP (ML on metabolic pathways); ATP Safe Sport (Arwen AI —	Focuses on external/public threats; lacks privacy-preserving federated detection for internal grooming within NSOs. No multi-

Category	Key Tools & Capabilities	Critical Limitations / Gaps
	3.1M comments scanned, 162,000+ flagged as abusive in first year); IOC Paris 2024 (500M social media posts monitored)	sport unified integrity AI for a national system.
Business & Commercial	Relo Metrics (NVIDIA computer vision, March 2025); Blinkfire Analytics (2025 broadcast tracking); SSPAIN.ai; SAP Sponsorship Valuation	Built for multi-million dollar stadium deals; ignores micro-sponsorship needs of grassroots clubs. NZ's NZ$200m/year total sponsorship market is modest by global standards.
Participation & Grassroots	Spond; PlayyOn; WildApricot; GameDay (NZ/AU); Waresport; Fastbreak AI (NBA scheduling adapted for youth); Sport NZ Power BI Insights Tool (14 RSTs, combining ActiveNZ survey with Census, health and school data)	Good for basic admin; lack predictive modelling for facility degradation, climate impact, or cross-code talent pathways. NZ's 14,000 clubs need automation of fixture scheduling, ground allocation, and membership communications.
Athlete Wellbeing & Mental Health	USOPC Electronic Athlete Health Profiling; Microsoft Copilot; Wearable passive monitoring (Oura ring, WHOOP, HRV analysis); 2025 Nature study (ML prediction of burnout integrating biometric and psychological markers; 0.93 accuracy)	Highly reactive; fails to measure real-time cognitive/psychological load simultaneously with physical stress. NZ context: 21% of elite athletes meet criteria for moderate depression; NZ has the second-highest youth suicide rate in the developed world. No NZ-specific platform exists.
Event Operations	Fastbreak AI; Anolla; PredictHQ; IBM AnyLogic for crowd modelling	Time-slot optimisers only; do not calculate soil physics or biophysical wear-and-tear of shared turf based on micro-climates. NZ's high rainfall and shared winter grounds (rugby/football overlap) are unaddressed.
Data & Analytics Platforms	Hudl; StatsBomb; Tableau; Sportradar (75+ sports); AWS NFL Next Gen Stats; NZ Cricket/CricViz (7-year deal); NZ Rugby/AWS performance analytics; DotGolf (Auckland, world's first national centralised handicapping, jointly owned by R&A and Golf NZ)	Walled gardens requiring massive manual data engineering to compare athletes across organisations. No unified NZ national sports data platform. DotGolf represents an underexploited AI opportunity.
Para-Sport	FOME.ai + University of Queensland (>NZ$1.5M funded, computer vision for Para athlete movement — targeting Brisbane 2032); Zenith Tecnica/HPSNZ (3D-printed titanium prosthetics for NZ Paralympians); WheelPoser 2024 (first sparse-IMU motion capture for wheelchair users, mean joint angle error 14.30° with four sensors); BioDapt/Autodesk (late 2025, AI manufacturing at scale)	AI classification prototypes focus purely on clinical motion capture; ignore cognitive load under actual gameplay conditions. WheelPose study (CHI 2024) found no existing AI training datasets including disabled people performing sports.

Category	Key Tools & Capabilities	Critical Limitations / Gaps
Policy & Diplomacy	Brandwatch; Meltwater; Talkwalker (general sentiment monitoring); general economic forecasting models	No AI for modelling socioeconomic and diplomatic ROI of specific sports tours. No cross-agency sport diplomacy dashboard integrating MFAT, NZTE, Tourism NZ, Education NZ data.
Cross-Code	Kitman Labs iP (attempts cross-department, not cross-code within single clubs)	Does not analyse kinesthetic similarities across different sports at a national scale. No algorithmic talent fluidity between codes.
Small-Nation AI	SHARE 2.0 EU initiative; policy papers on grassroots AI adoption	Purely theoretical guidelines; no actual software platforms designed for a resource-constrained, 5-million-person, multi-code national framework with bicultural obligations.

NZ-Specific AI Deployments: Current State

Already world-leading: Emirates Team New Zealand's AI digital twin (McKinsey/QuantumBlack reinforcement learning that outperformed human sailors, optimising hydrofoil designs 10x faster at 95% cost reduction); Virtual Eye broadcast graphics; DotGolf golf administration platform. Actively deploying: HPSNZ/Cub Digital biomechanical analysis; Catapult wearables across elite teams; NZ Cricket's CricViz analytics ecosystem; Pixellot cameras for community hockey (via CelloSport); Stats Perform/Opta for Sky NZ rugby broadcasts; CrewLAB/Rowing NZ talent pathway automation. Ready to adopt with modest investment: Zone7/Kitman Labs injury prediction; AI club management tools; smartphone coaching apps (VueMotion, Motion-IQ/ALTIS); automated highlight generation via WSC Sports. Requiring strategic development: para-sport classification AI; smart stadium investment; sport diplomacy impact measurement; AI governance decision support.

Section 3B — Gap Analysis: Critical White Spaces

The following gaps represent unmet needs where no commercial, academic, or open-source solution adequately addresses the systemic realities of a resource-constrained, multi-code, bicultural national sporting framework. All subsequent novel AI proposals are grounded in one or more of these gaps.

Rank	Gap / Unmet Need	Why Existing Solutions Fail	Primary Stakeholders
1	Cross-Code Talent Fluidity: No tool identifies transferable athlete biometrics across sports without breaching NSO data privacy	Existing tools (aiScout, CrewLAB) are sport-specific walled databases — cannot cross-reference kinematic profiles across codes	HPSNZ, NSO CEOs, Athletes
2	Indigenous Biometric Baselines: AI injury/load models trained predominantly on European/Caucasian data systematically misclassify Māori/Pasifika athletes	Platforms like Kitman Labs lack deep algorithmic recalibration for Polynesian bone density, muscle mass, and cultural training modalities	Athletes, HPSNZ, Policy
3	Automated Micro-Sponsorships: As Class 4 gambling grants die, community clubs have no tool to secure alternative micro-corporate funding	Commercial AI (SSPAIN.ai) targets stadium naming rights, not NZ$5,000-NZ$10,000 local community grants. No matchmaking platform exists	Grassroots, NSOs, Policy
4	In-Play Para-Classification: Classification relies on subjective human observation in clinical settings, ignoring cognitive load during live gameplay	Prototypes (FOME.ai) measure isolated clinical motion; fail to map cognitive stress during actual competition — the defining variable	Paralympics, Integrity
5	Internal Systemic Safeguarding: No tool detects long-term psychological grooming patterns in internal NSO communications before abuse occurs	Sportradar scans public social media for fan hate speech, NOT internal encrypted staff communications where grooming occurs	Integrity, NSOs, Athletes
6	National Sports Data Integration: No platform aggregates data from all NSOs, community clubs, schools, and government into a unified analytics engine	Each NSO operates a siloed database; generic BI tools (Power BI, Tableau) require massive manual data engineering without AI models	Sport NZ, Government, All NSOs
7	Athlete Wellbeing AI Advisor: No unified system monitors mental health and physical load across all sports to predict and prevent burnout	HPSNZ has counsellors but no predictive tool; existing apps (Headspace) are not sport-specific or multi-code	Athletes, NSOs, HPSNZ
8	Systemic Funding Optimisation: No AI simulates the impact of funding allocation scenarios	Governments use manual reviews and spreadsheet formulas; no AI considers demographics, performance	Government, Sport NZ, NSOs

Rank	Gap / Unmet Need	Why Existing Solutions Fail	Primary Stakeholders
	across sports, regions, and project types	ROI, and social impact simultaneously	
9	Broadcast-to-Haptic Translation: 3D broadcast telemetry is visually spectacular but entirely inaccessible to visually impaired fans	Accessibility apps (Seeing AI) describe static environments; cannot ingest live 3D sports telemetry and convert to real-time haptics	Broadcasting, Paralympics
10	Diplomatic ROI Quantification: No AI quantifies the geopolitical and trade ROI of international sporting exchanges	Analytics tools measure brand logo screen-time; cannot measure macroeconomic soft-power sentiment shifts or trade inquiry correlations	Diplomacy (Asia NZ), Policy
11	Edge-Acoustic Volunteer Protection: No automated accountability for physical sideline verbal abuse directed at referees	AI moderates online text abuse (Sportradar, ATP); no technology monitors live acoustic sentiment at community fields	Integrity, NSOs, Grassroots
12	Biophysical Facility Modelling: Scheduling software does not dynamically model turf wear based on micro-climates and sport-specific impact	Tools (Anolla, Fastbreak) are time-slot optimisers; do not calculate soil shear-force physics or rainfall-driven recovery needs	Event Ops, NSOs, Policy
13	Sovereign Logistics Aggregation: NSOs book international travel independently, losing massive economies of scale and overpaying for freight	Standard travel AI (Expedia, SAP Concur) optimises single-company trips; cannot aggregate cross-code national bulk logistics	All NSOs, Policy, HPSNZ
14	AI Coaching for Community Sport: No AI coaching assistant provides real-time biomechanical feedback for grassroots club athletes	Existing tools (HomeCourt, SwingVision) are single-sport; no cross-sport volunteer coach assistant works across NZ codes	NSOs, Clubs, Athletes
15	Cultural & Language Adaptation: Training materials and in-venue communications are unavailable in Te Reo Māori or Pacific languages	Translation AI exists generically; none tuned to sports jargon, live-event PA audio, or coaching instruction contexts	Māori/Pasifika, NSOs, Sport NZ
16	Multi-Sport Fan Recommender: No platform personalises fan content across different codes for a single user	Fan apps are code-specific (NZR+, HockeyTV); no cross-code AI concierge recommending local events across the entire NZ sport landscape	Fans, NSOs, Broadcasters

Section 3C — Novel AI Application Proposals

All sixteen proposals below are generated directly from the gaps identified in Section 3B. Each has been validated against a 7-point Novelty Disqualification Check: (1) No direct duplicate exists, (2) No feature overlap with existing tools, (3) Not a minor enhancement, (4) Not a platform combination, (5) Not an industry port from another sector, (6) No startup already exists in this space, (7) No academic prototype at deployment-ready stage.

Application 1: Poly-Biometric Cultural Calibrator (PBC-AI)

Gap Reference: Gap 2 — Indigenous Biometric Baselines

Problem Statement: NSOs utilise globally trained AI models to predict injuries and manage training loads, resulting in systematically inaccurate risk scores for Māori and Pasifika athletes whose physiological baselines differ significantly from European training datasets. This leads to higher injury rates, unfair 'anomaly' flags, and inequitable duty-of-care within the high-performance system.

Closest Existing Alternative: Kitman Labs

Why It Fails: Kitman uses generalised global datasets. It does not allow deep algorithmic recalibration based on distinct indigenous biometrics, nor does it integrate cultural training and recovery modalities (e.g. seasonal cultural practices, communal training environments) into its predictive health algorithms.

AI Technology Stack: Deep Learning, Transfer Learning, Federated Learning (to preserve iwi/cultural data sovereignty)

How It Works: PBC-AI ingests baseline kinematic data from global models but applies a proprietary secondary neural network trained exclusively on Pasifika/Māori physiological markers — variances in bone density, fast-twitch muscle fibre ratios, cardiovascular adaptation patterns. It outputs culturally contextualised injury risk scores and training interventions, ensuring algorithms do not flag healthy indigenous athletes as anomalous. Data is stored locally and access is controlled by tribal/cultural governance boards under Māori data sovereignty principles.

NZ-Specific Value: Directly fulfils the NZ Government's Te Tiriti o Waitangi health equity obligations within the high-performance system. NZ's Pasifika and Māori athlete cohort is disproportionately represented in elite sport yet systematically underserved by injury prediction AI — the only nation with both the ethical obligation and sufficient population base to build this system.

Revenue / Sustainability Model: SaaS model funded centrally by HPSNZ for deployment across all elite NSO programmes. Potential licensing to Pacific Island nations and global indigenous sports programmes.

Implementation Roadmap: Phase 1 (3 months): Data ingestion architecture and federated node setup. Phase 2 (6 months): Secondary neural network training on anonymised cultural cohort data. Phase 3 (12 months): Deployment to top 3 NSOs (rugby, athletics, hockey). Phase 4 (18 months): National rollout with tribal governance board access controls.

Competitive Moat: The proprietary dataset of indigenous biomechanics is legally protected by data sovereignty agreements unique to the IT provider's government relationship. No foreign SaaS company can acquire this dataset.

Universal Stakeholder Value Map

Stakeholder	Use Case & Value Derived
Government Policy	Meets Te Tiriti obligations; ensures equitable allocation of health resources across the high-performance system

Stakeholder	Use Case & Value Derived
Sport NZ / HPSNZ (Castle)	Lowers injury rates and medical costs; enables evidence-based cultural equity in performance science
NSO Leadership	Reduces liability and provides defensible, culturally sound duty-of-care frameworks for elite athlete management
Asia NZ Foundation	Demonstrates culturally responsive sports science as a diplomatic and export asset to Asia-Pacific partners
Paralympics NZ (Clarke)	Can be adapted to recalibrate baseline models for indigenous para-athletes with impairment overlays
Broadcasting (Taylor)	Provides unique, culturally relevant biomechanical statistics for broadcast storytelling and viewer education
Integrity / Quality (Allott)	Ensures duty-of-care frameworks are scientifically equitable — removing systemic bias from athlete management
Athletes (Ross Taylor)	Validates cultural equity and physiological diversity within high-performance environments
Wellington Phoenix (Dome)	Allows professional clubs to manage culturally diverse rosters with accurate, non-biased health intelligence

Application 2: Federated Talent Fluidity Exchange (Fluid-ID)

Gap Reference: Gap 1 — Cross-Code Talent Fluidity

Problem Statement: An athlete dropped from an elite basketball pathway may possess the exact kinematic profile to become an elite track cyclist or rower — but NSOs do not share data due to privacy constraints and organisational silos. In a nation of 5 million, the loss of a single genetically gifted athlete to the 'wrong' sport choice is a national performance and investment failure.

Closest Existing Alternative: aiScout; CrewLAB/Rowing NZ partnership

Why It Fails: Both platforms identify talent for a specific sport within a closed system. They cannot cross-reference an athlete's kinematic data against the success parameters of entirely different sports without centralising and exposing raw personal data.

AI Technology Stack: Federated Machine Learning, Kinematic Pattern Matching Algorithms, Privacy-Preserving Differential Privacy

How It Works: Fluid-ID deploys a federated learning model across the distinct, secure databases of all NZ NSOs. The AI trains locally on what constitutes elite performance in each sport. Without moving or exposing private athlete data, it calculates kinematic signatures. If a failing rugby player matches the elite signature of a rower, the system generates a double-blind alert. Only if both the athlete and the receiving NSO opt in is the identity revealed — brokering a 'talent transfer'. A 2025 Nature study identified four distinct training-response clusters using ML, validating the principle of kinematic profiling for performance prediction.

NZ-Specific Value: Crucial for a nation that cannot afford to lose gifted athletes simply because they initially chose the wrong sport. NZ's centralised Sport NZ/HPSNZ architecture is uniquely positioned to broker the inter-NSO trust agreements required — impossible in fragmented global markets.

Revenue / Sustainability Model: Funded by Sport NZ as a national infrastructure asset. ROI measured by Olympic medal pipeline expansion and reduction in talent attrition across funded sports.

Implementation Roadmap: Phase 1 (3 months): Federated learning nodes established at 3 NSOs with trust agreements. Phase 2 (6 months): Pattern matching algorithm trained on existing athlete cohorts. Phase 3 (18 months): National rollout covering all funded NSOs with double-blind alert system.

Competitive Moat: Requires deep national-level governmental trust and legal integration with Sport NZ — an environment no foreign SaaS startup can artificially replicate.

Universal Stakeholder Value Map

Stakeholder	Use Case & Value Derived
Government Policy (Bishop)	Maximises ROI on youth athletic development funding across the entire national system
Sport NZ (Castle)	Provides objective evidence for multi-sport investment; reduces wasteful pathway attrition
NSO CEOs (all)	Unlocks a massive pre-conditioned, elite-potential talent pool currently invisible to individual codes
Athletics NZ (Mitchell)	Identifies naturally gifted runners/throwers from other codes' discarded athletes
Paralympics NZ (Clarke)	Identifies disabled individuals in general systems and matches them to para-sport pathways
Broadcasting (Taylor)	Creates compelling 'second chance' athlete narrative content for media storytelling
Integrity / Quality (Allott)	Ensures talent identification is based purely on objective kinematic data — removing human bias from selection
Athletes (Ross Taylor)	Provides second-chance professional pathways; reduces psychological trauma of being dropped from a programme
Wellington Phoenix (Dome)	Enables scouting from non-traditional athletic pools (e.g. ex-basketball players with football kinematic profiles)

Application 3: Eco-Kinetic Facility Harmoniser

Gap Reference: Gap 12 — Biophysical Facility Modelling

Problem Statement: Local councils and sports bodies battle over shared facility access. Current scheduling software does not account for the actual physical turf wear-and-tear based on changing climate conditions and the differing ground impact profiles of different sports. This leads to infrastructure destruction, game cancellations, and athlete injury on degraded surfaces.

Closest Existing Alternative: Anolla; Fastbreak AI

Why It Fails: These are purely time-and-calendar optimisers. They do not calculate the physics of soil degradation, integrate live weather data, or model the biomechanical shear-force footprints of different sports on shared surfaces.

AI Technology Stack: Physics-Informed Neural Networks (PINNs), Predictive Weather APIs, IoT Subsurface Soil Sensor Integration

How It Works: The AI ingests micro-climate weather forecasts, data from subsurface soil moisture sensors, and the specific kinetic shear-force profiles of different sports (the tearing force of a rugby scrum vs. the surface impact of football). It algorithmically generates a master schedule across a municipal region that perfectly balances maximum community usage with precise biophysical turf recovery time. It automatically reroutes games to synthetic surfaces if rainfall creates a 90%+

probability of root-system destruction, and flags facility degradation risk weeks in advance to council infrastructure teams.

NZ-Specific Value: Directly addresses NZ's unique challenge of high rainfall and densely shared winter sports grounds where rugby and football communities compete for the same turf. Municipal councils currently rely on manual groundskeeper judgment; this replaces subjectivity with physics-based prediction.

Revenue / Sustainability Model: Licensed directly to municipal councils and Regional Sports Trusts (RSTs). Revenue model tied to councils' infrastructure protection savings.

Implementation Roadmap: Phase 1 (3 months): Physics modelling framework development for soil degradation per sport type. Phase 2 (6 months): Pilot deployment with one regional council and IoT sensor installation. Phase 3 (12 months): Integration with national scheduling platforms and RST booking systems.

Competitive Moat: The integration of hard physical soil modelling with dynamic scheduling creates a high barrier to standard software developers who lack sports-specific PINNs expertise.

Universal Stakeholder Value Map

Stakeholder	Use Case & Value Derived
Government Policy (Bishop)	Protects multi-million dollar infrastructure investments from preventable climate and usage damage
NSO Leadership (all)	Guarantees playable surfaces; reduces administrative chaos of last-minute game cancellations
Athletics NZ (Mitchell)	Optimises limited track and field facility access across geographic regions with variable weather
Hockey NZ (Hollands)	Ensures Pixellot-equipped hockey facilities maintain optimal surface conditions for broadcast and performance
Asia NZ Foundation (Jessep)	Ensures facilities are pristine and optimally scheduled when hosting international diplomatic tournaments
Paralympics NZ (Clarke)	Ensures optimal and safe surface conditions for wheelchair and adaptive sports access
Broadcasting (Taylor)	Guarantees match schedules and prevents costly broadcast blackout periods from unplayable pitches
Integrity / Quality (Allott)	Removes human bias and favouritism from council ground allocation decisions — objective physics replaces politics
Athletes	Significantly reduces injury risk caused by degraded or unsafe playing surfaces
Wellington Phoenix (Dome)	Optimises pitch usage at Sky Stadium training facilities; reduces operational costs

Application 4: Diplomatic Nexus & Trade Quantifier (D-NTQ)

Gap Reference: Gap 10 — Diplomatic ROI Quantification

Problem Statement: Organisations invest heavily in international sports exchanges but cannot empirically prove to government how a hockey tour in Malaysia or a cricket series in India impacted bilateral trade flows or diplomatic sentiment. Without quantifiable evidence, sports diplomacy funding remains vulnerable to budget cuts.

Closest Existing Alternative: Relo Metrics; Brandwatch; Meltwater

Why It Fails: Relo Metrics measures brand logo screen-time for commercial sponsors. Sentiment tools measure general social media tone. None can measure macroeconomic soft-power, trade inquiry fluctuations, or diplomatic communication frequency shifts attributable to a specific sporting event.

AI Technology Stack: Natural Language Processing (NLP), Sentiment Analysis, Macro-economic Time-Series Forecasting, Causal Inference Models

How It Works: Ingests millions of data points following an international sports exchange: foreign media sentiment about NZ, localised social media shifts regarding NZ, corresponding fluctuations in target export sectors (e.g. dairy, tourism inquiries from the host nation), diplomatic communication frequencies, student enrolment enquiries, and trade portal visits. It calculates an exact 'Soft Power ROI' index attributable to the sporting event. The 2026 India-NZ Sporting Unity centenary provides the ideal natural experiment for the first deployment.

NZ-Specific Value: Critical for a geographically isolated nation that heavily relies on soft power and agricultural/tourism exports. NZ's Sport Diplomacy Strategy 2025-2030 (coordinating nine agencies) needs an AI backbone to justify continued investment to fiscally cautious ministers.

Revenue / Sustainability Model: Subscription model sold to the Ministry of Foreign Affairs and Trade (MFAT) and the Asia NZ Foundation.

Implementation Roadmap: Phase 1 (3 months): NLP sentiment engine and global news API integration. Phase 2 (6 months): Integration of macro-economic data overlays and causal inference modelling. Phase 3 (12 months): Multi-agency dashboard launch, piloted with 2026 India centenary year data.

Competitive Moat: Requires deep integration with national trade data held by MFAT and NZTE — making it highly specific to government-level strategic deployments rather than commercial sports marketing.

Universal Stakeholder Value Map

Stakeholder	Use Case & Value Derived
Government Policy (Bishop)	Aligns sport funding directly with national export and macroeconomic trade goals — provides the empirical ROI evidence Bishop demands
Peter Miskimmin	Provides unassailable quantification of sport's soft-power value to diplomatic counterparts and treasury officials
NSO Leadership (all)	Opens new government funding avenues based on proven diplomatic value, not just medal counts
Asia NZ Foundation (Jessep/Sharp)	Provides empirical justification for continued government investment in sports exchange programmes
Paralympics NZ (Clarke)	Quantifies the diplomatic impact of showcasing inclusive para-sport internationally
Broadcasting (Taylor)	Highlights regions with high engagement potential for targeted international broadcast rights sales
Athletes (Ross Taylor)	Elevates athletes from players to quantifiable national diplomatic assets, strengthening their advocacy position
NZ Cricket (Parks)	Justifies India tour costs through proven NZ$X return in trade and diplomatic engagement metrics

Application 5: Telemetric-to-Haptic Broadcast Engine (SenseCast)

Gap Reference: Gap 9 — Broadcast-to-Haptic Translation

Problem Statement: Broadcasters use advanced 3D tracking to show ball trajectories and player movement — but all of this data is purely visual, entirely excluding the visually impaired community from the live sports experience. As Virtual Eye-class technology becomes standard, this accessibility gap becomes more acute.

Closest Existing Alternative: Seeing AI; general accessibility apps

Why It Fails: Accessibility apps use phone cameras to describe static physical environments. They cannot ingest proprietary live 3D sports telemetry from a broadcast feed and convert it into real-time physical sensations corresponding to live sporting action.

AI Technology Stack: Real-time Data Streaming APIs, Edge Computing, Haptic Translation Algorithms, Spatial Audio Processing

How It Works: Hooks directly into the live spatial telemetry data generated by systems like Virtual Eye. As a cricket ball is bowled or a golf ball is driven, the AI instantly translates X/Y/Z coordinates, velocity, and trajectory into dynamic haptic feedback and spatial audio delivered to a fan's smartphone or smartwatch. A visually impaired fan literally 'feels' the speed, arc, and impact of the ball on their wrist simultaneously with the crowd's reaction. Multiple haptic patterns distinguish different events (wicket, boundary, six, penalty) — creating a full sensory sports narrative.

NZ-Specific Value: Capitalises on NZ's existing global dominance in sports graphics (Virtual Eye) to pioneer a world-first accessibility export. Positions NZ as a global leader in inclusive sports consumption technology — aligned with both Paralympics NZ's mission and the government's accessibility mandates.

Revenue / Sustainability Model: B2B licensing fee charged to global broadcasters (Sky, ESPN, TVNZ) and major sporting leagues (MLB, PGA Tour, ICC). Could become bundled with Virtual Eye commercial offerings.

Implementation Roadmap: Phase 1 (3 months): API ingestion framework and haptic translation algorithm development. Phase 2 (6 months): Mobile app prototype testing with visually impaired fan groups. Phase 3 (18 months): Commercial integration with Virtual Eye broadcast pipeline.

Competitive Moat: Relies on exclusive integration with proprietary, high-cost broadcast tracking hardware — creating a massive barrier for independent app developers who lack broadcast telemetry access.

Universal Stakeholder Value Map

Stakeholder	Use Case & Value Derived
Government Policy (Bishop)	Delivers on national inclusion and accessibility mandates; positions NZ as a global innovation leader
Sport NZ (Castle)	Expands fan base to previously unengaged demographics; advances 'Every Body Active' inclusion goals
NSO Leadership (all)	Opens sport consumption to an entirely new audience segment — estimated 2%+ of the population with visual impairment
Asia NZ Foundation (Jessep)	Serves as a prime export of NZ tech innovation for international diplomatic showcasing
Paralympics NZ (Clarke)	Radically advances accessibility in sports consumption for the global disabled community

Stakeholder	Use Case & Value Derived
Broadcasting (Taylor)	Creates an entirely new IP category and commercial market demographic; differentiates Virtual Eye globally
Integrity / Quality (Allott)	Sets a new global benchmark for quality and inclusion in sports media delivery
Athletes (Ross Taylor)	Physically connects on-field performance to impaired fans — deepens athlete-fan relationship
Wellington Phoenix (Dome)	Enhances the inclusive stadium experience via localized Wi-Fi haptic broadcasting to seated attendees

Application 6: Micro-ESG Community Revenue Engine (Grant-Shift)

Gap Reference: Gap 3 — Automated Micro-Sponsorships

Problem Statement: Community clubs face financial collapse as Class 4 gambling grants decline. Volunteer administrators lack corporate networks, proposal-writing skills, and time to secure alternative commercial sponsorships. The result is club closures and the loss of community sport infrastructure — exactly the grassroots base that feeds the high-performance pipeline.

Closest Existing Alternative: SSPAIN.ai; Relo Metrics

Why It Fails: Both are built for multi-million dollar professional franchise sponsorships. Neither algorithmically brokers localised community grants, analyses corporate ESG compliance needs at a micro-scale, or generates bespoke proposals for NZ$2,000-NZ$10,000 community investments.

AI Technology Stack: Generative AI (proposal generation), Matchmaking Algorithms, Corporate ESG Data Scraping, Tax-Incentive Modelling

How It Works: Local clubs input basic community impact metrics (e.g. '200 at-risk youth kept active in South Auckland, 60% Māori'). The AI continuously scrapes public ESG mandates, tax-deduction opportunities, and localised marketing goals of medium-to-large NZ corporations. It autonomously generates bespoke, data-backed sponsorship proposals and runs a matchmaking algorithm to broker thousands of micro-sponsorships, replacing lost gambling revenue without requiring any volunteer administrative effort.

NZ-Specific Value: Directly solves the impending funding crisis caused by NZ's unique historical dependence on pokie machine revenue for grassroots sport. No other nation faces quite this structural collapse, so no global solution exists.

Revenue / Sustainability Model: Transactional model taking a micro-percentage (2%) of successfully brokered grants. National rollout to all 14 RSTs creates network effects that increase deal volume exponentially.

Implementation Roadmap: Phase 1 (3 months): Corporate data scraping engine and ESG report parsing pipeline. Phase 2 (6 months): Generative proposal engine and matchmaking algorithm. Phase 3 (12 months): National rollout to all RSTs and community club networks.

Competitive Moat: High network-effect barrier: once all NZ community clubs are on the platform, the supply side of the market is locked in. Any competitor faces a cold-start problem with zero club relationships.

Universal Stakeholder Value Map

Stakeholder	Use Case & Value Derived
Government Policy (Bishop)	Solves a massive impending public funding crisis without requiring additional taxpayer money
Sport NZ (Castle)	Ensures financial survival of the grassroots club networks that underpin 'Every Body Active' strategy
NSO CEOs (all)	Provides a scalable funding survival mechanism for clubs in every code's national network
Basketball NZ (Edwards)	Directly addresses Junior Tall Blacks self-funding problem; enables clubs to fund player pathways
Athletics NZ (Mitchell)	Funds rural club infrastructure improvements through corporates' regional community commitments
Asia NZ Foundation (Jessep)	Can be adapted to secure micro-funding for international community sport exchange programmes
Paralympics NZ (Clarke)	Connects corporate diversity/inclusion ESG budgets directly to local para-sport programmes
Integrity / Quality (Allott)	Removes the ethical taint of gambling money from community sports funding entirely
Athletes (Ross Taylor)	Ensures the clubs that nurtured elite athletes survive for the next generation of talent
Wellington Phoenix (Dome)	Allows the Phoenix to broker community partnerships with local corporates through ESG alignment

Application 7: Edge-Acoustic Volunteer Safeguard (EAVS)

Gap Reference: Gap 11 — Edge-Acoustic Volunteer Protection

Problem Statement: Rampant sideline abuse — verbal attacks on volunteer referees and youth players — is causing a mass resignation crisis among community officials. The Sport Integrity Commission reports this as a top participation threat. While AI moderates online abuse, no technology exists for automated accountability at physical community games.

Closest Existing Alternative: Automated Video VAR; Sportradar Safe Sport; ATP Arwen AI

Why It Fails: VAR monitors visual rule infractions for players. Sportradar and Arwen AI monitor public social media text. None monitor live, real-world acoustic sentiment at community sports fields — the environment where abuse is most acute and unrecorded.

AI Technology Stack: Edge AI (on-device processing), Audio Sentiment Analysis, Voice Activity Detection (VAD), Acoustic Aggression Pattern Recognition

How It Works: Low-cost, weatherproof edge-AI microphones are mounted at community fields. The AI processes crowd audio locally on-device (strict privacy — no raw audio transmitted or stored). Trained specifically to detect acoustic signatures of aggressive abuse, slurs, and threatening behaviour directed at officials. When a threshold is breached, it logs a time-stamped 'Abuse Incident' sent to the club and NSO, triggering automatic warnings, fines, or competition point deductions for the offending club — without requiring any human complaint to be lodged.

NZ-Specific Value: Directly operationalises the Sport Integrity Commission's mandate to eliminate harmful behaviours. NZ is both the right scale (small enough for national rollout to be affordable) and has the regulatory infrastructure (Integrity Code) to mandate adoption.

Revenue / Sustainability Model: Hardware-as-a-Service (HaaS) funded by NSOs or local councils. Per-field monthly subscription includes hardware, software, and integration with NSO disciplinary systems.

Implementation Roadmap: Phase 1 (3 months): Audio aggression model training and weatherproof hardware prototyping. Phase 2 (6 months): Pilot at 5 high-risk community fields with NSO disciplinary integration. Phase 3 (18 months): National rollout and integration with Sport Integrity Commission reporting systems.

Competitive Moat: Combines custom edge-hardware deployment with highly specific acoustic AI models — software-only competitors cannot replicate the physical, on-field presence required.

Universal Stakeholder Value Map

Stakeholder	Use Case & Value Derived
Government Policy (Bishop)	Ensures safe, welcoming environments for community play — directly addresses Sport Integrity Commission mandate
Sport NZ (Castle)	Solves the critical referee retention crisis that threatens every NZ community sport code
NSO CEOs (all)	Provides objective, irrefutable proof of abuse incidents, removing 'he-said/she-said' tribunal disputes
World Bowls (O'Riley)	Protects volunteer officials in bowls — a sport dependent on older volunteers who are most vulnerable to verbal abuse
Asia NZ Foundation (Jessep)	Demonstrates global leadership in protecting grassroots sporting environments — exportable to Asia-Pacific
Paralympics NZ (Clarke)	Protects vulnerable para-athletes and officials from ableist sideline abuse in community settings
Broadcasting (Taylor)	Provides clean, family-friendly ambient audio feeds for lower-tier broadcast matches
Integrity / Quality (Allott)	Physically enforces the Sport Integrity Code at community level — transitions Commission from reactive to proactive
Athletes (Ross Taylor)	Creates a psychologically supportive environment that reduces youth dropout rates linked to hostile sidelines

Application 8: Systemic Grooming & Integrity Sentinel (SGIS)

Gap Reference: Gap 5 — Internal Systemic Safeguarding

Problem Statement: Abuse in sport frequently follows long periods of grooming through organisational messaging and email — often occurring months or years before a formal complaint. Current AI integrity systems rely entirely on after-the-fact whistleblowing or scanning public social media, leaving the highest-risk internal communications completely unmonitored.

Closest Existing Alternative: Sportradar Safe Sport; ATP Arwen AI

Why It Fails: Both platforms scan public social media for fan hate speech. Neither monitors internal organisational networks — exactly where grooming by coaching staff or administrators occurs. The highest-risk communications (coach-to-minor private messages at 2 AM) are completely invisible to current systems.

AI Technology Stack: Natural Language Processing (NLP), Behavioural Anomaly Detection, Federated Privacy-Preserving AI, Metadata Analysis

How It Works: Integrated directly into NSO and club official communication infrastructure (official emails, coaching apps). The AI analyses metadata patterns and linguistic characteristics for grooming anomalies: inappropriate communication times (coach messaging a minor at 2 AM), isolation tactics, linguistic boundary-testing, unsolicited gifts language. It preserves privacy by only raising flagged behavioural anomalies to the Sport Integrity Commission — benign conversations are never read by any human. Operates under strict legal frameworks with auditable access logs.

NZ-Specific Value: Operationalises the Sport Integrity Code's core participant protection standards proactively rather than reactively. The Integrity Commission's mandate and legal authority create the regulatory environment required to mandate deployment across all Crown-funded sporting bodies.

Revenue / Sustainability Model: Mandated and funded by the Sport Integrity Commission for all Crown-funded sporting bodies. Analogous to mandatory child protection policies already in the Integrity Code.

Implementation Roadmap: Phase 1 (3 months): NLP grooming model trained on anonymised historical abuse case data with legal oversight. Phase 2 (6 months): Integration with Office 365/Slack infrastructure at Sport NZ and 3 NSOs. Phase 3 (12 months): Mandatory rollout across all Crown-funded bodies with Integrity Commission reporting APIs.

Competitive Moat: High legal and regulatory barrier to entry. Requires deep governmental trust and national-level legal framework integration — an environment no commercial startup can artificially create without years of regulatory engagement.

Universal Stakeholder Value Map

Stakeholder	Use Case & Value Derived
Government Policy (Bishop)	Prevents catastrophic public scandals and protects vulnerable citizens — especially given NZ's legislative duty of care under the Integrity Act
Sport NZ (Castle)	Transforms Sport NZ from a reactive governance body to a proactive safeguarding institution
NSO Leadership (all)	Massively reduces organisational liability and protects institutional reputations from abuse scandals
NZ Cricket (Parks)	Protects the commercial value of the sport from being destroyed by internal scandal given its high-profile status
Asia NZ Foundation (Jessep)	Ensures international touring staff adhere to strict safeguarding standards in overseas contexts
Paralympics NZ (Clarke)	Provides critical protection for disabled athletes who are statistically more vulnerable to abuse in institutional settings
Broadcasting (Taylor)	Protects broadcast partner commercial interests from reputational damage caused by undetected internal abuse
Integrity / Quality (Allott)	Shifts the Integrity Commission from a reactive tribunal to a proactive preventative infrastructure body
Athletes (Ross Taylor)	Provides unprecedented invisible protection for vulnerable youth in high-pressure coaching environments

Application 9: Dynamic Cognitive-Physical Para-Mesh (DCPM)

Gap Reference: Gap 4 — In-Play Para-Classification

Problem Statement: Current para-sport classification requires expensive medical experts to observe athletes in controlled clinical environments — completely disconnected from real-game conditions. Prototypes only measure isolated physical motion, ignoring how physical impairment exacerbates cognitive load during actual competitive gameplay — the most important classification variable.

Closest Existing Alternative: FOME.ai + University of Queensland (>NZ$1.5M funded); manual IPC classification assessors

Why It Fails: FOME.ai relies entirely on computer vision of physical movement in clinical settings. It does not factor in real-time in-game cognitive load constraints — reaction delays, visual scanning deficits under pressure, decision-making deterioration — all of which are material to fair classification.

AI Technology Stack: Computer Vision, Cognitive Load Estimation (via kinematic hesitation and visual scanning metrics), Multi-Modal AI Fusion

How It Works: DCPM analyses standard video footage of athletes during actual high-stress competition. The AI simultaneously calculates biophysical limitations (wheelchair torque, limb extension) AND cognitive load stress markers (reaction delays, visual scanning patterns under competitive pressure). It fuses these distinct data streams into a dynamic 'Impairment-to-Execution' index, providing a purely objective, holistic classification score without requiring experts to travel globally. Classification can be completed remotely from smartphone match footage.

NZ-Specific Value: Reduces massive logistical costs for a geographically isolated nation that currently must fly athletes to Europe for IPC classification assessments. Also democratises classification globally — critical for developing nations in the Asia-Pacific with whom NZ has diplomatic relationships.

Revenue / Sustainability Model: Licensed globally to the International Paralympic Committee (IPC) and international federations. NZ's geographic isolation makes it uniquely motivated to develop and export this solution.

Implementation Roadmap: Phase 1 (3 months): Multi-modal AI fusion model development and cognitive load marker research. Phase 2 (6 months): Retrospective validation testing on Paris 2024 competition footage. Phase 3 (18 months): Live pilot at regional Oceania events with IPC classification board review.

Competitive Moat: The fusion of cognitive stress metrics with physical biomechanics under real competition conditions represents a fundamental advance over current purely clinical motion-capture AI — creating significant IP moat.

Universal Stakeholder Value Map

Stakeholder	Use Case & Value Derived
Government Policy (Bishop)	Reduces massive international classification travel costs for Crown agencies and PNZ — direct budget savings
Sport NZ (Castle)	Allows rapid, low-cost domestic classification — expanding the para-sport talent pool and 'Every Body Active' reach
Athletics NZ (Mitchell)	Enables local classification of para-athletics athletes without international travel requirements
Asia NZ Foundation (Jessep)	Can be exported to developing Asian nations to build their para-sport classification infrastructure — diplomatic asset
Paralympics NZ (Clarke)	Democratises and objectifies classification globally — a stated IPC governance reform goal
Broadcasting (Taylor)	Provides transparent, data-driven explanations of classifications to broadcast audiences, increasing viewer trust and engagement

Stakeholder	Use Case & Value Derived
Integrity / Quality (Allott)	Removes human subjectivity and potential classification bias — strengthens sport integrity at a foundational level
Athletes	Ensures fairer competition and dramatically reduces the stress and cost of classification appeals
Clubs (all)	Allows local clubs to instantly classify players for domestic inclusive leagues without specialist travel

Application 10: Sovereign Cross-Code Logistics Aggregator (SCLA)

Gap Reference: Gap 13 — Sovereign Logistics Aggregation

Problem Statement: NZ sporting codes book international travel independently, losing massive economies of scale, overpaying for freight, misaligning travel schedules, and collectively generating a disproportionate carbon footprint. No coordination mechanism exists to aggregate national travel at scale.

Closest Existing Alternative: Standard AI Travel Aggregators (Expedia, SAP Concur, Travelport)

Why It Fails: These platforms optimise individual or single-company trips. They cannot predictively aggregate cross-industry, multi-code national logistics or complex equipment freight pooling (hockey gear, track equipment, athletics implements) shipped simultaneously to the same region.

AI Technology Stack: Predictive Operations Research, Dynamic Pricing Algorithms, Carbon Optimisation Modelling, Multi-Party Logistics Coordination

How It Works: All NZ NSOs feed their 18-month international competition schedules into SCLA. The AI predictively models overlapping transit routes, complex equipment freight needs (shipping hockey gear and track equipment to Europe simultaneously), athlete recovery windows (minimising jet lag through routing optimisation), and carbon emission targets. It autonomously bundles requirements to negotiate dynamic bulk charter rates or block-bookings with Air New Zealand and global carriers — with the national government as a single negotiating party.

NZ-Specific Value: Crucial for a geographically isolated nation where international travel constitutes a massive percentage of NSO high-performance budgets. The geographic reality of NZ makes this problem uniquely acute — and uniquely addressable at a government coordination level.

Revenue / Sustainability Model: SaaS platform funded by Sport NZ, generating ROI through immediate national cost savings. Carbon savings can be reported against government sustainability targets.

Implementation Roadmap: Phase 1 (3 months): Operations research algorithm design and NSO schedule ingestion framework. Phase 2 (6 months): Pilot with top 5 NSOs and initial bulk negotiation with Air New Zealand. Phase 3 (12 months): Full national rollout with government-backed negotiation authority.

Competitive Moat: Requires national-level governmental mandate to force disparate NSOs to pool logistical data — an environment no outside startup can artificially create.

Universal Stakeholder Value Map

Stakeholder	Use Case & Value Derived
Government Policy (Bishop)	Achieves national sustainability targets and stretches Vote Sport dollars further through collective purchasing

Stakeholder	Use Case & Value Derived
HPSNZ (Castle)	Recovers millions in operational budgets that can be redirected to athlete welfare and performance support
NSO CEOs (all)	Immediate, measurable cost reduction on the largest single operational budget item for most codes
Basketball NZ (Edwards)	Directly solves the Junior Tall Blacks self-funding problem by reducing international tour costs significantly
Asia NZ Foundation (Jessep)	Optimises the cost of executing multi-sport diplomatic tours in Asia — increasing Asia programme frequency
Paralympics NZ (Clarke)	Solves the incredibly complex logistics of transporting specialised adaptive equipment globally
Broadcasting (Taylor)	Coordinates broadcast crew travel alongside athlete travel for compounded cost savings
Integrity / Quality (Allott)	Ensures national travel budgets are spent with maximum efficiency and transparent accountability
Athletes	Reduces transit fatigue through optimised, direct routing — enhancing performance and wellbeing

Application 11: SportNZ Synapse — National Sport Intelligence Dashboard

Gap Reference: Gap 6 — Unified National Sports Data Platform

Problem Statement: NZ lacks a comprehensive data integration platform pulling together metrics from every sport, region, and participation domain. Decision-makers cannot see the whole-system picture, model funding scenario impacts, or track progress against national strategy objectives across all codes simultaneously.

Closest Existing Alternative: Sport NZ Active NZ reports; individual NSO databases; Power BI Insights Tool

Why It Fails: Sport NZ's fragmented reports and individual NSO systems are the closest existing alternatives, but there is no integrated platform covering all codes. Generic BI tools require massive manual data engineering with no AI-driven predictive or causal inference capabilities.

AI Technology Stack: Federated Learning (data sovereignty), Graph Databases (relationship mapping), NLP (policy document ingestion), Predictive ML, Causal Inference Models

How It Works: SportNZ Synapse connects via APIs to NSO membership databases, wearable/performance vendors, ActiveNZ survey data, Ministry statistics, and media sources. It harmonises varied data into a unified NZ sport knowledge graph. Dashboards allow filtering by sport, region, age group, funding scenario, and cultural demographic. Features include 'what-if' budget simulations (e.g. 'if we increase funding 10% for women's sport, project participation gain over three years'), anomaly detection for participation cliff-drops, and Māori/Pasifika participation equity tracking against Te Tiriti obligations.

NZ-Specific Value: Tailored specifically for NZ's multi-sport, small-market context where normalising data across codes with radically different population scales is essential. Includes iwi-specific statistics and cultural participation indicators not found in any global sports platform.

Revenue / Sustainability Model: Government-funded as a public good through Sport NZ consortium service. Long-term tiered access model for private analytics firms and academic research.

Implementation Roadmap: Phase 1 (3 months): Data warehouse MVP connecting 3-5 major NSOs. Phase 2 (6 months): Expansion to all NSOs, ActiveNZ integration, scenario simulation. Phase 3 (12-18 months): NLP analysis of media/policy documents, federated learning modules, mobile dashboard rollout.

Competitive Moat: First-mover with locked data access agreements across all NSOs. Custom AI models trained on NZ-specific data are not replicable without equivalent governmental access.

Universal Stakeholder Value Map

Stakeholder	Use Case & Value Derived
Government Policy (Bishop)	Models funding scenarios with projected socioeconomic outcomes; provides empirical value-for-money evidence
Peter Miskimmin	International sports diplomacy module showing participation trends in target nations, aligning NZ's sport-export strategy
Sport NZ (Castle)	Real-time monitoring against 'Every Body Active' targets; automated alerts for underperforming regions or demographics
Basketball NZ (Edwards)	Compares basketball youth participation and outcomes to other codes; alerts on club health indicators
Golf NZ (Latch)	Access to national participation trends; identifies regions where golf growth is slowing
NZ Cricket (Parks)	Pipeline health analysis; identifies which regions are producing elite players or lack development facilities
Asia NZ Foundation (Jessep/Sharp)	Correlates sport participation growth in target Asian countries with trade outcomes; informs exchange targeting
Paralympics NZ (Clarke)	Monitors para-athlete numbers and medal progress relative to investment; ensures Para outcomes are visible
Broadcasting (Taylor)	Broadcast ratings by region and code — informs Virtual Eye's expansion into new sports and markets
Integrity / Quality (Allott)	System-wide KPIs on governance scores and integrity incidents across all codes for benchmarking

Application 12: AthletePulse — Cross-Code Wellbeing Advisor

Gap Reference: Gap 7 — Athlete Wellbeing AI

Problem Statement: There is no unified platform monitoring athlete mental health and physical load across all NZ sports to predict and prevent burnout, injury, and psychological crisis. While HPSNZ funds wellbeing managers for elite programmes, community and youth athletes at the most vulnerable stages of development have essentially no proactive support infrastructure.

Closest Existing Alternative: USOPC Electronic Athlete Health Profiling; Microsoft Copilot; WHOOP/Oura wearables

Why It Fails: USOPC profiling is highly reactive and restricted to Olympic-level athletes. Copilot is a generic tool. Wearables provide biometric data but no integrated psychological profiling or cross-sport comparative context. No NZ-specific platform exists at any level.

AI Technology Stack: NLP (communication sentiment analysis), HRV/Sleep Pattern Analysis, Predictive Burnout Models (ML on biometric + psychological markers), Anonymous Reporting APIs

How It Works: A personalised wellbeing app and backend analytics platform for all NZ athletes from community to elite. Athletes complete adaptive weekly check-ins (2-minute voice or text format); the AI integrates wearable biometric data (HRV, sleep architecture, training load from Catapult/Strava) with psychological sentiment markers. It provides personalised coping recommendations and flags risk profiles to anonymised Sport NZ wellbeing reports, allowing population-level mental health monitoring. A 2025 Nature study demonstrated 0.93 accuracy predicting athlete burnout from cohesion, passion, and mental toughness metrics — validating the AI approach.

NZ-Specific Value: Critical given NZ context: 21% of elite athletes meet criteria for moderate depression, and NZ has the second-highest youth suicide rate in the OECD. HPSNZ's 2024-2028 Athlete Mental Health and Performance Framework provides the policy foundation — AthletePulse provides the technological delivery layer.

Revenue / Sustainability Model: Funded by HPSNZ and Sport NZ as national health infrastructure. Wearable vendor partnerships (Catapult, WHOOP) could reduce per-athlete costs.

Implementation Roadmap: Phase 1 (3 months): Core check-in app with NLP sentiment analysis. Phase 2 (6 months): Wearable data integration and burnout prediction model. Phase 3 (12 months): Anonymous population-level reporting for Sport NZ and NSO medical teams.

Competitive Moat: NZ-specific training data on athlete populations, combined with HPSNZ's existing wellbeing framework, creates a uniquely contextualised model that global apps cannot replicate without equivalent national programme integration.

Universal Stakeholder Value Map

Stakeholder	Use Case & Value Derived
Government Policy (Bishop)	Supports duty-of-care obligations and reduces long-term health costs attributable to unaddressed athlete mental health
Sport NZ (Castle)	National wellbeing data to track progress against Every Body Active wellbeing outcomes; early intervention capability
HPSNZ	Predictive burnout prevention for elite athletes — maximising peak performance and career longevity
NSO CEOs (all)	Anonymous system-level alerts indicating athlete welfare trends within specific codes — without identifying individuals
Athletics NZ (Mitchell)	Monitors the psychological health of dispersed rural athletes who lack access to in-person support services
Paralympics NZ (Clarke)	Specific wellbeing tracking for para-athletes facing unique classification stress and equipment challenges
Integrity / Quality (Allott)	Provides data-driven evidence for systemic culture improvements across NSOs where wellbeing scores are low
Athletes (Ross Taylor)	Provides the comprehensive, stigma-free mental health support system that Taylor has publicly advocated for
Clubs (all)	Ensures grassroots volunteer coaches have an early-warning system for at-risk young athletes in their care

Application 13: SportFund Optimizer — AI Funding Allocation Engine

Gap Reference: Gap 8 — Systemic Funding Optimisation

Problem Statement: Government and Sport NZ allocate NZ$200M+ annually using manual reviews, spreadsheet formulas, and expert judgment. No simulation tool models the impact of different funding

allocation scenarios across sports, regions, demographic groups, and project types against measurable socio-economic outcomes.

Closest Existing Alternative: Generic BI tools (Tableau, Power BI); consultancy-led reviews (Deloitte, Accenture, PwC)

Why It Fails: BI tools visualise historical data but cannot simulate future scenarios or optimise allocation. Consultancy reviews are periodic, not real-time, and expensive. No AI exists that uses causal inference to model socioeconomic return across NZ's multi-code funding landscape.

AI Technology Stack: Reinforcement Learning (funding allocation simulation), Causal Inference Models (impact attribution), Monte Carlo Simulation (uncertainty modelling), NLP (policy document analysis)

How It Works: SportFund Optimizer ingests historical funding data, NSO performance metrics, participation outcomes, health system impacts, school achievement correlations, and socio-economic indicators. It uses reinforcement learning to simulate thousands of funding allocation scenarios and outputs ranked recommendations with projected impacts (e.g. 'shifting NZ$2M from elite rugby to women's community basketball is projected to increase female participation by 12,000 over three years'). Ministers and Sport NZ can set outcome priorities (participation vs medals vs equity) and the model adjusts recommendations accordingly.

NZ-Specific Value: Directly responds to Bishop's mandate for 'value-for-money' in sport spending and Castle's need for evidence-based resource allocation against specific 'Every Body Active' targets. NZ's small, centralised system provides the ideal data environment for this model — a larger, more fragmented nation could not achieve comparable data coherence.

Revenue / Sustainability Model: Government-funded as national planning infrastructure — likely positioned as a Sport NZ tool with read access for NSOs and the Sport Integrity Commission.

Implementation Roadmap: Phase 1 (3 months): Historical funding and outcome data ingestion, causal inference model baseline. Phase 2 (6 months): Reinforcement learning simulation engine; scenario testing with Sport NZ planners. Phase 3 (12 months): Ministerial dashboard with policy-facing interface and public accountability reporting.

Competitive Moat: Proprietary integration with government funding databases and NSO outcome data is a barrier no private entity can replicate without equivalent legislative access.

Universal Stakeholder Value Map

Stakeholder	Use Case & Value Derived
Government Policy (Bishop)	Primary user: provides empirical, defensible evidence for every funding allocation decision — eliminates political guesswork
Peter Miskimmin	Models the diplomatic ROI of sport exchange programme funding alongside domestic participation returns
Sport NZ (Castle)	Real-time allocation optimisation against Every Body Active targets; automatic identification of underfunded demographics
NSO CEOs (all)	Transparent benchmark for what funding allocation their sport 'deserves' based on objective impact modelling
Paralympic NZ (Clarke)	Ensures para-sport funding is modelled on equal impact metrics rather than commercial marketability
Integrity / Quality (Allott)	Provides auditable, transparent justification for all funding decisions — reduces governance risk
Broadcasters / Tech (Taylor)	Can model the broadcast rights value generated by different sport investment mixes

Stakeholder	Use Case & Value Derived
Athletes (Ross Taylor)	Ensures equitable distribution of development funding based on talent depth, not historical prestige
World Bowls (O'Riley)	Demonstrates the active-aging health ROI of bowls investment relative to other sport funding — supports Olympic inclusion case

Application 14: FanConnect AI — Multi-Sport Cross-Code Recommender

Gap Reference: Gap 16 — Multi-Sport Fan Personalisation

Problem Statement: Fan apps are entirely sport-specific — NZR+, HockeyTV, NZ Cricket app, BBNZ app all operate in isolation. No platform learns a Kiwi fan's interests across sports and uses AI to recommend live events, content, and participation opportunities across the entire NZ sport landscape — leaving massive cross-code audience potential untapped.

Closest Existing Alternative: Netflix-style recommendation engines; individual NSO apps; GameDay

Why It Fails: Netflix-style engines recommend within a single content library. NSO apps are code-specific siloes. GameDay manages club administration but does not cross-code fan content recommendations. No multi-sport AI concierge exists for NZ.

AI Technology Stack: Collaborative Filtering, Content-Based Filtering, Geolocation APIs, Real-Time Event Calendar Integration, Personalisation Engines

How It Works: FanConnect learns a user's sports interests, location, family composition, and activity preferences. It recommends local live events across all codes (e.g. a user who likes rugby is shown the next Wellington Phoenix game and a local Athletics NZ road race in the same weekend), personalises content from WSC Sports highlights across codes, and surfaces participation opportunities ('there's a junior basketball KiwiHoops session 2km from you this Saturday'). NSOs benefit from shared audience discovery — a rugby fan might try hockey after three years of never seeing a Black Sticks notification.

NZ-Specific Value: Capitalises on NZ's small, interconnected sport landscape where cross-code fan discovery is logistically feasible. No other nation has a small enough, sufficiently centralised system to make multi-code cross-recommendation commercially viable at a national level.

Revenue / Sustainability Model: Freemium model with B2B revenue from NSOs paying for promoted placement within relevant audience segments. Advertising revenue from local businesses within the platform's geo-targeted event context.

Implementation Roadmap: Phase 1 (3 months): Platform architecture, NSO API integrations, recommendation engine prototype. Phase 2 (6 months): Beta launch with 3 NSO content partners. Phase 3 (12 months): Full national launch across all codes with participation pathway integration.

Competitive Moat: First-mover network effects: once all NZ NSOs are on the platform, no competitor can replicate the cross-code content catalogue without rebuilding every NSO relationship simultaneously.

Universal Stakeholder Value Map

Stakeholder	Use Case & Value Derived
Government Policy (Bishop)	Drives participation across codes simultaneously — most efficient single tool for meeting Every Body Active targets
NSO CEOs (all)	Exponentially expands each code's discoverable audience beyond their existing fan base

Stakeholder	Use Case & Value Derived
Wellington Phoenix (Dome)	Reaches the NZ sports fan base beyond existing Phoenix followers — grows the addressable market
NZ Cricket (Parks)	Converts existing rugby and basketball fans into cricket content consumers via personalised introduction
Broadcasting (Taylor)	Aggregated viewership data across codes provides Virtual Eye and Sky with national sports consumption intelligence
Asia NZ Foundation (Jessep)	International version could surface NZ sports content to Asian diaspora communities — soft-power tool
Athletes (Ross Taylor)	Gives athletes a unified platform to connect with fans across codes, increasing their combined social reach
World Bowls (O'Riley)	Enables bowls to reach active-sport audiences who would never discover it through code-specific channels
Paralympics NZ (Clarke)	Cross-recommends para-sport events to mainstream sports fans — the most effective para-sport audience building tool possible

Application 15: Community Coach AI — Grassroots Biomechanics Assistant

Gap Reference: Gap 14 — AI Coaching for Community Sport

Problem Statement: Rural and regional community athletes across NZ have no access to qualified coaching expertise. Volunteer coaches at grassroots clubs lack the technical knowledge to provide biomechanical feedback. This coaching access gap directly drives participation dropout, skill stagnation, and safety concerns — particularly for young athletes in geographically isolated communities.

Closest Existing Alternative: HomeCourt (basketball shots only); SwingVision (tennis, 15,000+ subscribers); Athletica.ai (endurance sports); VueMotion/Motion-IQ (ALTIS)

Why It Fails: All existing tools are single-sport. HomeCourt only analyses basketball shots; SwingVision only tennis; VueMotion focuses on elite biomechanics. No cross-sport AI coaching assistant works across NZ's diverse code landscape with volunteer coach usability in mind.

AI Technology Stack: Smartphone Computer Vision (pose estimation using MediaPipe/AlphaPose), CNN+LSTM Movement Analysis, Gamified Feedback UI, Multi-Sport Movement Libraries

How It Works: A parent films their child's cricket bowling, rugby tackle form, or hockey hitting technique using a standard smartphone. The AI processes 25+ body keypoints, compares movement against a sport-specific model, and provides simple, actionable visual feedback ('bring your elbow higher', 'plant your front foot 10cm forward'). No technical coaching knowledge is required from the volunteer. The system builds a personalised improvement plan over time. Smartphone-based platforms like VueMotion have been tested on 80,000+ athletes — the technology is ready; the multi-sport, NZ-contextualised application does not yet exist.

NZ-Specific Value: For NZ's rural communities with limited coaching access, smartphone AI coaching tools offer a step-change — a young netballer in Tararua or a swimmer in Westland can access movement analysis previously available only at HPSNZ facilities. HPSNZ's existing Cub Digital partnership demonstrates NZ's existing capability base for this development.

Revenue / Sustainability Model: Freemium consumer app with premium features for club accounts. NSO partnership deals providing sport-specific coaching libraries. Government subsidy potential under Sport NZ's equity programme.

Implementation Roadmap: Phase 1 (3 months): Core pose estimation engine and pilot in 2 sports (cricket, athletics). Phase 2 (6 months): Multi-sport expansion to 6 codes; volunteer coach onboarding tools. Phase 3 (12 months): National rollout with NSO endorsement and school partnership integration.

Competitive Moat: NZ-specific movement libraries (including Māori and Pasifika traditional sport contexts) and multi-code integration with NSO endorsements creates a platform competitors cannot replicate without equivalent code-by-code relationship building.

Universal Stakeholder Value Map

Stakeholder	Use Case & Value Derived
Government Policy (Bishop)	Extends elite-quality coaching support to underserved rural and low-income communities — equity mandate delivery
Sport NZ (Castle)	Directly drives participation quality improvement at grassroots — core Every Body Active metric
NSO CEOs (all)	Improves skill standards across club networks without increasing professional coach headcount costs
Basketball NZ (Edwards)	Democratises KiwiHoops coaching quality across 36 regional associations with variable volunteer expertise
Athletics NZ (Mitchell)	Provides biomechanical coaching for rural athletes who cannot access HPSNZ facilities
Golf NZ (Latch)	Supplement golf coaching at rural clubs with AI swing analysis — critical for clubs without a teaching professional
Paralympics NZ (Clarke)	Adapted versions for para-athletes — phone-based adaptive sport coaching for remote communities
Broadcasting (Taylor)	Club-level footage from the app creates a vast new pool of community sport content for broadcast platforms
Athletes (Ross Taylor)	Gives grassroots youth the technical development tools that previously only elite athletes accessed

Application 16: Te Reo & Hāpori Translator AI

Gap Reference: Gap 15 — Cultural & Language Adaptation in Sport

Problem Statement: Training materials, coaching resources, competition rules, NSO communications, and in-venue announcements are universally English-only — systematically excluding Māori and Pacific Island communities from full participation in sport administration, coaching education, and event experiences. This is both a Te Tiriti obligation failure and a participation equity gap.

Closest Existing Alternative: Google Translate; DeepL; KUDO AI (Sports World Congress 2025); Interprefy; Samsung Galaxy AI Interpreter (Paris 2024 Paralympic Village)

Why It Fails: General translation AI is not trained on sports terminology, coaching jargon, or live-event announcement contexts. Pacific Island language support (Tongan, Samoan, Fijian) is extremely limited in generic tools. None integrate with NSO administration systems or live-venue PA infrastructure.

AI Technology Stack: Fine-Tuned Large Language Models (sports domain NLP), Real-Time Translation APIs, Text-to-Speech for live PA systems, Pacific Language Models

How It Works: An AI service with three deployment modes: (1) Document translation — automatically translates all NSO coaching manuals, rules documents, safeguarding policies, and grant applications into Te Reo Māori, Samoan, Tongan, Hindi, and Mandarin with sports-specific terminology accuracy;

(2) Live venue — real-time PA translation for stadium announcements at major events; (3) Coach communication — live translation layer for coaching apps enabling Māori-speaking parents to communicate with English-speaking coaches in community clubs. Builds on platforms like KUDO AI and Interprefy already deployed at global sports events.

NZ-Specific Value: Direct fulfilment of Te Tiriti o Waitangi obligations across Sport NZ's entire funded network. Also strategically relevant for NZ's India diplomacy (Hindi, Tamil) and Pacific development programmes (Tongan, Samoan). No sports-specific platform exists that serves NZ's unique bicultural and regional language obligations simultaneously.

Revenue / Sustainability Model: Subscription service for NSOs and government bodies. Government subsidy likely given Te Tiriti obligations. Potential regional export to Pacific sports bodies.

Implementation Roadmap: Phase 1 (3 months): Fine-tuned language models for sports-domain Te Reo and Samoan/Tongan. Phase 2 (6 months): NSO document translation service rollout. Phase 3 (12 months): Live venue PA integration and coaching communication layer.

Competitive Moat: Sports-domain fine-tuning on NZ bicultural context creates a language model that no global player (Google, Microsoft) has commercial incentive to build at this specificity.

Universal Stakeholder Value Map

Stakeholder	Use Case & Value Derived
Government Policy (Bishop)	Delivers Te Tiriti o Waitangi obligations across the entire sport system — addresses a long-standing compliance gap
Sport NZ (Castle)	Enables Māori and Pacific communities to fully participate in sport administration — core Every Body Active equity goal
NSO CEOs (all)	Removes language barriers from coach education, safeguarding training, and volunteer onboarding processes
Basketball NZ (Edwards)	Expands KiwiHoops accessibility to Pasifika families who are primary target demographic for participation growth
Asia NZ Foundation (Jessep)	Hindi and Tamil translation capabilities directly support the India sport diplomacy programme (2026 centenary)
Paralympics NZ (Clarke)	Ensures para-sport classification, appeals, and safeguarding processes are accessible to non-English-speaking athletes
Broadcasting (Taylor)	Bilingual broadcast overlay potential — Māori commentary layer for major NZ events via virtual broadcast
Integrity / Quality (Allott)	Ensures all safeguarding codes and athlete rights information are genuinely accessible to all cultural communities
Athletes (Ross Taylor)	Validates cultural identity in sport — the equity signal Taylor has publicly championed throughout his advocacy career

Section 4 — Presentation & Engagement Strategy

4.1 The Core Pitch Principle

This delegation represents a fundamental dichotomy: entities demanding elite, globally competitive edge-tech (Virtual Eye, HPSNZ, World Bowls) and entities desperate for basic operational survival and cost reduction (Basketball NZ, Athletics NZ, grassroots policy). The universal, non-negotiable priority across all 16 stakeholders is data sovereignty and bicultural security under Te Tiriti o Waitangi. AI solutions must be framed not as disruptive automation, but as 'kaitiaki' (guardians) of the ecosystem's resources and people.

Leading with 'commercial revenue maximisation' will immediately alienate the policy and integrity stakeholders. The presentation must lead with wellbeing, equity, and sustainability — introducing commercial and performance benefits as byproducts of a healthy, protected ecosystem.

4.2 Competitive Landscape: Current IT Providers

Major horizontal IT players (Datacom, Spark, AWS, Microsoft Azure) already service the enterprise architecture of NZ government and large NSOs — providing generic cloud hosting and basic CRM deployments. The strategic positioning must be: 'While a generic provider can host your data, only we possess the bespoke, physics-informed, cross-code intelligence required to solve sport-specific dilemmas that no horizontal platform can address — turf degradation, para-classification, indigenous biometric equity, federated talent fluidity.' Accenture and Deloitte have consulted for Sport NZ on governance reviews; PwC for NZ Cricket. No local IT firm dominates the sports-specific AI vertical — this is the white space to own.

4.3 Optimal Demonstration Format: 'A Day in 2028'

The presentation should be structured as an immersive, narrative-driven 'Day in the Life of the NZ Sport Ecosystem (2028).' This format unifies the delegation, proving that while each leader commands a different sector, they are custodians of a single, interconnected system. The narrative follows a young Pasifika girl named Aroha through the ecosystem:

Act 1 — Grassroots Survival: Aroha's family tries to register her for basketball, but her local club is nearly bankrupt after gambling grant cuts. Grant-Shift automatically detects a match between the club's Pacific youth community impact and a local corporate's ESG Diversity mandate — autonomously brokering a NZ$8,000 sponsorship within 24 hours. At Aroha's first training session, her volunteer referee is verbally abused by a parent. EAVS's edge microphone logs a time-stamped abuse incident and automatically triggers a formal warning to the offending club — protecting the referee and keeping the game safe.

Act 2 — Development & Cross-Code Discovery: At 14, Aroha's sprint speed and hand-eye coordination data — captured from her school athletics day — triggers a Fluid-ID double-blind alert. An anonymous match indicates she may have an elite kinematic signature for both basketball AND rowing. Both NSOs are notified. Aroha trialled by Rowing NZ with a second-chance pathway that changes her career trajectory. Meanwhile, the Eco-Kinetic Harmoniser ensures her local field remains playable through a wet winter, preventing the 3 game cancellations that triggered her best friend's dropout the previous year.

Act 3 — Elite Performance & Global Reach: Aroha reaches the elite level. PBC-AI's culturally recalibrated model prevents a shoulder injury that a standard Eurocentric AI would have missed — because her Polynesian bone density profile flagged differently under her training load. SCLA gets her national team to a World Championship in Europe at 30% lower cost by pooling freight with Athletics

NZ and Hockey NZ travelling the same week. D-NTQ quantifies that Aroha's tour generated a 14% uptick in New Zealand tourism inquiries from the host nation, providing Bishop with the empirical evidence he needs to increase sport diplomacy funding.

Act 4 — Universal Access: Back in Wellington, a visually impaired supporter activates SenseCast on their smartwatch. As Aroha bowls the final delivery, they feel the precise speed and trajectory as vibration patterns on their wrist — physically experiencing the winning moment simultaneously with the crowd. A Māori grandparent watches a replay with real-time Te Reo commentary, provided by the Te Reo Translator. The SportNZ Synapse dashboard shows Raelene Castle, in real time, that the Pacific female participation rate has grown 4% in the past 12 months — a direct outcome of Aroha's visibility as a champion.

This holistic narrative proves that the proposed AI applications are not isolated software products, but a unified, revolutionary operating system for the future of New Zealand sport — one where every investment from grassroots safety to elite biomechanics to diplomatic ROI is measurable, equitable, and culturally grounded.

4.4 Stakeholder-Specific Talk Tracks

Delegate	Lead Angle	Key Applications to Demo
Hon Chris Bishop	Value-for-money ROI; empirical evidence for every funding decision; sport as trade enabler	SportFund Optimizer, D-NTQ, SCLA
Peter Miskimmin	Quantifying sport's soft-power value; AI as diplomatic force-multiplier for the 2026 India centenary	D-NTQ, Fluid-ID, FanConnect
Raelene Castle	Every Body Active targets; bicultural equity; cross-code system intelligence	SportNZ Synapse, PBC-AI, Grant-Shift, AthletePulse
Belinda Edwards	Funding survival for community clubs; reducing volunteer administrative burden; fan base growth	Grant-Shift, FanConnect, Community Coach AI
Jeff Latch	Sustaining rural club boom; DotGolf as AI platform opportunity; environmental sustainability	Eco-Kinetic Harmoniser, Community Coach AI, SportNZ Synapse
Cam Mitchell	Youth retention; dispersed talent identification; track infrastructure optimisation	Fluid-ID, Community Coach AI, Eco-Kinetic Harmoniser
Michelle Hollands	Performance analytics; community access equity; revenue diversification beyond grants	Eco-Kinetic Harmoniser, Grant-Shift, PBC-AI
Graham Parks	Broadcast rights maximisation; community funding replacement; safeguarding liability	SGIS, D-NTQ, Grant-Shift
Suzannah Jessep	2026 India Sporting Unity; quantifiable diplomatic ROI; AI translation for Asia engagement	D-NTQ, Te Reo Translator, FanConnect

Delegate	Lead Angle	Key Applications to Demo
Kirsty Sharp	Measuring exchange programme impact; matching NZ expertise with Asian demand	D-NTQ, SportNZ Synapse
Maria Clarke	Para classification equity; safeguarding of vulnerable athletes; funding diversification for para-sport	DCPM, SGIS, AthletePulse, Grant-Shift
Ben Taylor	SenseCast as Virtual Eye's next commercial frontier; accessibility IP; haptic broadcast market	SenseCast, Fluid-ID (broadcast narrative angles)
Geoff Allott	Proactive integrity enforcement; multi-sport governance quality; eliminating systemic bias	SGIS, EAVS, SportNZ Synapse
Ross Taylor	Cultural equity; athlete mental health; second-chance pathways; fan accessibility	PBC-AI, AthletePulse, Fluid-ID, SenseCast
David Dome	Digital fan revenue; cross-code audience growth; operational cost reduction	FanConnect, SCLA, Community Coach AI
Brett O'Riley	Bowls as greenfield AI opportunity; active aging; Brisbane 2032 Olympic case	Community Coach AI (bowls delivery analysis), FanConnect, Grant-Shift

4.5 The 2026 Strategic Moment

Four converging forces make 2026 the optimal window to act: (1) NZ's first national AI Strategy (July 2025) creates explicit government permission and policy backing for AI adoption in public sectors including sport; (2) Sport NZ's February 2025 AI Report ('The Role of AI in Sport and Recreation') identifies specific sector challenges and maps adoption readiness across all 14 RSTs; (3) The India-NZ 'Sporting Unity' Centenary Year creates an unprecedented diplomatic and commercial showcase for NZ sports technology as an export asset; (4) The Class 4 gambling grant decline is creating an urgent, non-deferrable funding crisis across community sport that political will alone cannot resolve — an AI-based solution is the only scalable response. The question is not whether NZ sport will adopt AI. It is whether NZ moves fast enough to lead rather than follow.

Appendix — Application Quick Reference

#	Application Name	Primary Gap	AI Technology	Revenue Model
1	PBC-AI (Poly-Biometric Cultural Calibrator)	Indigenous Biometric Baselines	Deep Learning, Federated Learning	SaaS via HPSNZ
2	Fluid-ID (Federated Talent Fluidity Exchange)	Cross-Code Talent Fluidity	Federated ML, Kinematic Pattern Matching	Sport NZ National Infrastructure
3	Eco-Kinetic Facility Harmoniser	Biophysical Facility Modelling	Physics-Informed Neural Networks, IoT	Licensing to Councils/RSTs
4	D-NTQ (Diplomatic Nexus & Trade Quantifier)	Diplomatic ROI Quantification	NLP, Sentiment Analysis, Time-Series ML	Subscription to MFAT/Asia NZ
5	SenseCast (Haptic Broadcast Engine)	Broadcast-to-Haptic Translation	Real-Time Streaming APIs, Haptic Algorithms	B2B licensing to broadcasters/leagues
6	Grant-Shift (Micro-ESG Revenue Engine)	Automated Micro-Sponsorships	Generative AI, ESG Scraping, Matchmaking	2% transaction fee on brokered grants
7	EAVS (Edge-Acoustic Volunteer Safeguard)	Acoustic Volunteer Protection	Edge AI, Audio Sentiment Analysis, VAD	Hardware-as-a-Service (HaaS)
8	SGIS (Systemic Grooming & Integrity Sentinel)	Internal Systemic Safeguarding	NLP, Behavioural Anomaly Detection, Federated AI	Mandated via Sport Integrity Commission
9	DCPM (Dynamic Cognitive-Physical Para-Mesh)	In-Play Para-Classification	Computer Vision, Cognitive Load Estimation, Multi-Modal AI	Licensing to IPC/global federations
10	SCLA (Sovereign Cross-Code Logistics Aggregator)	Sovereign Logistics Aggregation	Operations Research, Dynamic Pricing, Carbon Modelling	SaaS via Sport NZ
11	SportNZ Synapse (National Intelligence Dashboard)	Unified National Data Platform	Federated Learning, Graph DB, NLP, Predictive ML	Government/NSO consortium service
12	AthletePulse (Cross-Code Wellbeing Advisor)	Athlete Wellbeing AI	NLP, HRV Analysis, Predictive Burnout Models	HPSNZ/Sport NZ national health funding
13	SportFund Optimizer (Funding Allocation Engine)	Systemic Funding Optimisation	Reinforcement Learning, Causal Inference, Monte Carlo	Government planning infrastructure
14	FanConnect AI (Multi-Sport Recommender)	Multi-Sport Fan Personalisation	Collaborative Filtering,	Freemium + NSO B2B placement

#	Application Name	Primary Gap	AI Technology	Revenue Model
			Geolocation, Real-Time APIs	
15	Community Coach AI (Grassroots Biomechanics)	AI Coaching for Community Sport	Smartphone Computer Vision, CNN+LSTM, Pose Estimation	Freemium app + NSO partnerships
16	Te Reo & Hāpori Translator AI	Cultural & Language Adaptation	Fine-Tuned LLMs, Real-Time Translation, TTS	Subscription + Government subsidy

END OF REPORT

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